

Fuzzy Private Set Intersection from Density-Bounded Assumptions

Seunghun Paik[✉] and Jae Hong Seo^{*✉}

Department of Mathematics & Research Institute for Natural Sciences
Hanyang University, Seoul 04763, Republic of Korea
`{whitesoonguh, jaehongseo}@hanyang.ac.kr`

Abstract. We propose a new fuzzy private set intersection (FPSI) protocol, a two-party functionality that securely identifies similar items across private sets. Our design departs from the prevailing separation-based paradigm in existing constructions. Rather than enforcing common separation-based assumptions, e.g., “apart by 2δ ” or disjoint projection assumptions, we adopt a fundamentally different perspective: we explicitly allow collisions within δ -balls, but require that their multiplicity be bounded per axis. We formalize this by introducing a new structural assumption called *k-max collision*, which upper-bounds the number of items intersecting any δ -ball along each axis by a parameter k . This new assumption intrinsically reflects the axis-wise distinctness of the data through density, rather than pairwise separability. Following prior analyses of disjoint projection assumptions, we show the plausibility of *k-max collision* by proving that uniformly distributed data satisfies this property. Under the *k-max collision* assumption, we propose an FPSI protocol with computation/communication complexities scaling linearly with both parties’ set sizes and the data dimension for a fixed k . Our results demonstrate that separation assumptions are not necessary for efficient FPSI. Notably, our construction accommodates the intermediate-dimension regime ($1 \ll d \ll \kappa$ for the data dimension d and the statistical security parameter κ), which has remained neither efficiently supported nor well-justified under existing separation-based FPSI paradigms.

1 Introduction

For the universe set $\mathcal{U}$, let $X, Y \subset \mathcal{U}^d$ be two sets held by two parties. The fuzzy private set intersection (FPSI) is a two-party functionality that securely computes the set $\{(x, y) \in X \times Y : \mathbf{d}(x, y) \leq \delta\}$ for some distance metric $\mathbf{d}$ and the threshold $\delta > 0$. Due to its real-world applicability where privacy-preserving similarity matching is necessary, e.g., contact tracing [19] or biometric authentication [18], FPSI has been actively studied recently, and various protocol designs have been proposed [40, 13, 26, 27, 24, 2, 23, 3, 36, 39, 38].

However, the inherent fuzziness of the problem poses substantial challenges for efficient protocol design. To mitigate this difficulty, existing constructions typically impose structural assumptions on the input sets, restricting the structure

^{*} Corresponding author.

of set items. Among them, separation-based assumptions have been the dominant design choice, such as requiring items to be at least 2δ apart [26,27,24,2,36,38,3] or enforcing disjoint projections across axes [26,2,23,43,3]. These assumptions effectively control collisions within the δ -neighborhoods or their axis projections of set items, facilitating the design of fuzzy matching techniques such as spatial hashing [26] or fuzzy mapping [23].

While separation-based constructions have made impressive progress, their paradigms are complementary rather than overlapping. Protocols based on “apart by 2δ ” with spatial hashing incur $\Omega(2^d)$ overhead, rendering them practical only when d is a small constant [26,27,24,2,36,38,3]. In contrast, disjoint projection assumptions typically require $d > \kappa$ for the statistical security parameter κ [26,2,23,3,43]. This tension leaves a non-empty regime $1 \ll d \ll \kappa$ in which neither paradigm yields an efficient and well-justified construction.

The above observation motivates us to revisit the prevailing separation-based design paradigm, raising the following research question:

Are separation assumptions inherently necessary for efficient FPSI?

Such assumptions have been treated as necessary to avoid the quadratic overhead $O(MN)$ in the parties’ set size M, N [23,3]. Existing alternatives based on generic two-party computation techniques indeed avoid structural assumptions, but exhibit $O(MN)$ complexity [40,13,39,6]. More recently, a few studies have succeeded in eliminating these assumptions while avoiding $O(MN)$ overhead, but they suffer from worse asymptotic complexity $O((\log \delta)^d)$ [3].

1.1 Our Contribution

In this paper, we answer this question in the negative. We propose a novel efficient FPSI protocol for ℓ_∞ distance that does not rely on separation assumptions. Our key observation is that eliminating collisions in δ -balls or their projections may not be fundamental; bounding their multiplicity suffices. We formulate this boundedness of collision by devising a new structural assumption called *k-max collision* over the FPSI input sets. While this density-bounded assumption is fundamentally incomparable with separation-based assumptions¹, this assumption exhibits greater flexibility with respect to the data dimension. Under this assumption, our protocol achieves communication and computation complexities that scale linearly in both the set sizes and the dimension, while effectively accommodating the aforementioned intermediate region $1 \ll d \ll \kappa$. We summarize our contribution as follows.

Modeling δ -Closeness via Membership Tags We design our protocol by modeling the closeness test as a set intersection problem. Intuitively, for a set $X \subset \mathcal{U}^d$ and an element $y = (y_1, \dots, y_d) \in \mathcal{U}^d$, an element in X is δ -close to y if and only if their projection to any axis is δ -close. This implies that, for a set $S_{y_j, j}$ of the indices for X whose corresponding items are δ -close with y_j in j^{th}

¹ Fig. 3 shows examples where one assumption holds, but the other does not.

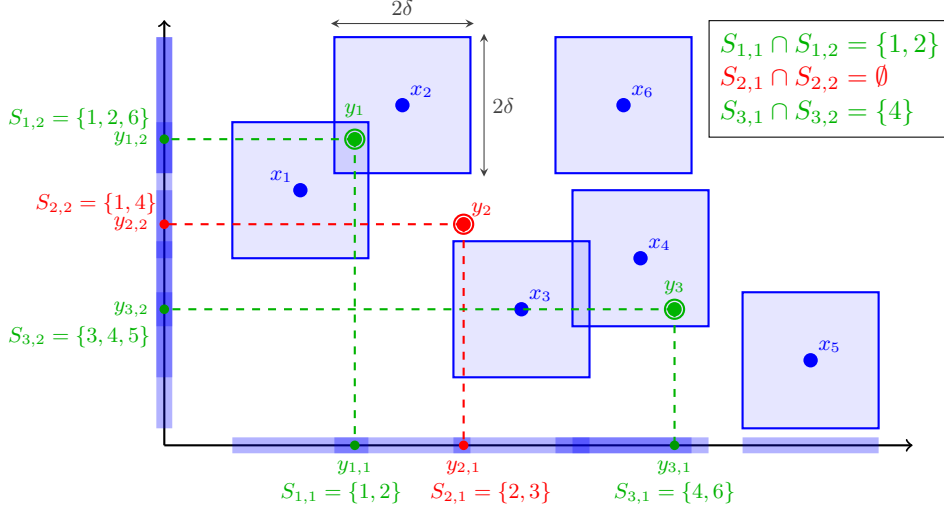


Fig. 1: Visualization of our approach on $X = \{x_1, \dots, x_6\}$ and $Y = \{y_1, y_2, y_3\}$ in $\mathcal{U}^2$. Each box depicts δ -neighborhoods of x_i in ℓ_∞ metric, and its projection for each axis is represented as bars. $S_{i,j}$ consists of indices of x_k whose δ -neighborhoods in j^{th} axis contains y_i for $i = 1, 2, 3$ and $j = 1, 2$. The membership of each y_i can be retrieved by checking whether $S_{i,1} \cap S_{i,2}$ is empty or not.

axis, then the items corresponding to indices in $\bigcap_{j=1}^d S_{y_j,j}$ are δ -close to y in $\mathcal{U}^d$. To help understand, we illustrate this axis-wise viewpoint in Fig. 1.

This observation motivates us to encode the set X into a collection of index sets $\{S_{v,j}\}$ for $j = 1, \dots, d$ and $v \in \mathcal{U}$, where $S_{v,j}$ is defined as the set of indices of X whose corresponding items are δ -close with v in j^{th} axis. Such an encoding represents the δ -closeness information of X ; we can retrieve the δ -close items in X by looking up corresponding $S_{v,j}$ for each axis and computing their intersection. We call these sets $S_{v,j}$ the *membership tags* of X .

The key benefit of this approach is that looking up $S_{y_j,j}$ from membership tags can be done coordinate-wise. Hence, unlike previous spatial hashing-based methods [26,27,24,2,36,38,3], we can avoid $\Omega(2^d)$ overhead. However, when utilizing this approach for designing an FPSI protocol, the key bottleneck is the size of each membership tag. Since each tag encodes the membership information of at most $M := |X|$ items, the tag size should be at least $\Omega(M)$. That is, to check the δ -closeness of a set Y over X , if we denote $N := |Y|$, at least $\Omega(MN)$ bits of information should be communicated. This implies that quadratic communication overhead seems unavoidable in this approach. Such overhead is inherited from the maximum number of indices encoded in each membership tag, and prior separation-based assumptions do not explicitly capture this quantity.

Efficient FPSI under k -Max Collision Assumption Our key observation is that the $\Omega(M)$ overhead in the tag size is inherited from the worst-case scenario, e.g., all set items are clustered in a narrow regime of a certain “bad” axis. In

contrast, in the average case where items are evenly spread over the axis, we can expect that the indices for each membership are *sparsely* distributed. Specifically, in the worst-case scenario mentioned above, we can consider such a “bad” axis as less informative, as it is unhelpful for discriminating between each item. That is, allocating $\Omega(M)$ bits to represent each membership tag would be overkill.

This observation suggests that what truly matters is not eliminating overlaps but bounding their multiplicity per axis. We formalize this bounded multiplicity as the k -max collision property, which enforces that the given set’s items overlap in their δ -neighborhood at most k times for each axis. For example, we can easily check that the set X in Fig. 1 has the 3-max collision property. In contrast to separation-based assumptions, this property explicitly allows the overlapping items in terms of their δ -neighborhoods. In our membership tag-based construction, however, these overlaps are admissible within a number bounded by the parameter k . If we assume that the set X we wish to encode is k -max collision, we can directly deduce that the membership tag size is at most k , i.e., $O(k)$, thus reducing the $\Omega(MN)$ overhead to $O(kN)$.

Under this assumption, we design an FPSI protocol based on membership tags by encoding each membership tag as a polynomial of degree at most k and realizing the lookup of tags via an oblivious key-value store [25]. We also instantiate the intersection of membership tags using standard polynomial-based techniques in the PSI literature [31], e.g., encrypting each membership tag polynomial via additive homomorphic encryption (AHE) and applying a random linear combination. We show that the resulting protocol achieves $O(kd\delta M + kN)$ communication and $O(k \log^2 k \cdot (d\delta M + N) + k^2 dN)$ computation complexity.

Supporting Intermediate Dimensions ($1 \ll d \ll \kappa$) To show the plausibility of the k -max collision assumption, we prove that this assumption holds for uniformly sampled set items over the universe with a dominating probability. In particular, for the data dimension d , we show that k scales with $\frac{\log d}{\log \log d}$ under the same failure probability. This property enables our protocol to exhibit different scaling behavior from protocols based on separation-based paradigms, which require $\Omega(2^d)$ complexity inherited from spatial hashing or a $d > \kappa$ condition to justify disjoint projection assumptions. More importantly, our protocol can seamlessly handle intermediate-dimensional regimes (e.g., $1 \ll d \ll \kappa$), while simultaneously supporting both the lower- (e.g., $d = O(1)$) and higher dimensional- (e.g., $d > \kappa$) regimes considered in previous studies. Concretely, for a parameter $(M, u, \delta) = (2^{16}, 32, 30)$ where $\mathcal{U} = [0, 2^u) \cap \mathbb{Z}$, which is typically considered the largest setting in many prior FPSI with ℓ_∞ metric [23, 2, 38, 43, 36, 3], $k = 7$ ($k = 8$, resp.) suffices to ensure the failure probability less than 2^{-40} for any $d \leq 262$ ($d \leq 2^{21}$, resp.).

1.2 Related Work

We survey existing fuzzy PSI protocol designs with respect to their design approach. Table 1 compares ours with existing FPSI protocols supporting ℓ_∞ metric in terms of supported dimension regimes and asymptotic complexity.

Table 1: Comparison of the FPSI protocols for ℓ_∞ metric on the universe $\mathcal{U} = [0, 2^u) \cap \mathbb{Z}$ and the threshold δ . We compare underlying assumptions, asymptotic communication/computation complexity, and the effective dimension regime d . The receiver and sender hold M and N in $\mathcal{U}^d$, respectively. We omit statistical/computational security parameters, and we merge the parties’ computational costs for readability. Factors that restrict the feasible range of d are highlighted in **red**. $\tilde{O}(\cdot)$ in ours means that $\text{poly}(\log k)$ factors are omitted.

Protocols	Assumption	Communication	Computation	Effective d
GRS22 [26]	$d_{\min} > 2\delta$	$O((4 \log \delta)^d M + N)$	$O((2\delta)^d M + (2 \log \delta)^d N)$	$d = O(1)$
GRS22 [26]	$d_{\min} > 4\delta$	$O(d 2^d \log \delta M + N)$	$O((2\delta)^d M + d \log \delta N)$	$d = O(1)$
GRS23 [27]	$d_{\min} > 8\delta$	$O(d \log \delta M + N)$	$O((2\delta)^d M + d \log \delta N)$	$d = O(1)$
vBP24 [2]	$d_{\min} > 2\delta$	$O(\delta d M + 2^d N)$	$O(\delta d M + d 2^d N)$	$d = O(1)$
vBP24 [2]	$d_{\min} > 4\delta$	$O(d 2^d \delta M + N)$	$O(d 2^d \delta M + d N)$	$d = O(1)$
DXL25 [17]	$d_{\min} > 4\delta$	$O(d 2^d \log \delta M + d \log \delta N)$	$O(d 2^d \log \delta M + d \log \delta N)$	$d = O(1)$
vBP25 [3]	$d_{\min} > 2\delta$	$O(d \log \delta (M + 2^d N))$	$O((\log \delta)^{d/2} M + d 2^d M \log \delta)$	$d = O(1)$
RRX24 [38]	disj. hash. ^α	$O(d \log \delta \cdot (2^d M + 2^{d-s} N))$	$O(d 2^d M + d \delta N)$	$d = O(1)$
PST ⁺ 25 [36]	disj. hash. ^α	$O(d 2^d M + d \delta M)$	$O(d 2^d M + d \delta N)$	$d = O(1)$
GGM24 [24]	disj. mini-univ. ^β	$O(d \log \delta M + (2 \log \delta)^d N)$	$O((\log \delta)^d M + (2 \log \delta)^d N)$	$d = O(1)$
BGMP25 [9]	disj. mini-univ. ^β	$O(d \log \delta M + (2 \log \delta)^d N)$	$O((\log \delta)^d M + (2 \log \delta)^d N)$	$d = O(1)$
GGM24 [24]	NA	$O(u d M + u^d M N)$	$O((u^d + u d) M N)$	$d = O(1)$
vBP25 [3]	NA	$O((M + N)(\log \delta)^d)$	$O((M + N)(\log \delta)^d)$	$d = O(1)$
GRS22 [26]	disj. proj. ^γ	$O(d \log \delta M + N)$	$O((2\delta)^d M + d \log \delta N)$	NA [†]
vBP24 [2]	disj. proj.	$O(d^2 \delta^2 M + N)$	$O(d^2 \delta^2 M + d^2 N)$	$d > \kappa$
GQL ⁺ 24[23]	disj. proj.	$O(d \delta M + d \delta N)$	$O(d \delta M + d \delta N)$	$d > \kappa$
ZCC ⁺ 25 [43]	disj. proj.	$O(\delta^s \frac{d}{s} (M + N))$	$O(\delta^s \frac{d}{s} (M + N))$	$d > \kappa$
DXL25 [17]	disj. proj.	$O(d \log \delta M + d \log \delta N)$	$O(d \log \delta M + d \log \delta N)$	$d > \kappa$
vBP25 [3]	disj. proj. ^γ	$O(d M + d \delta N)$	$O(d M + d \delta N)$	$d > \kappa$
Ours	k-Max Coll.	$O(k d \delta M + k N)$	$\tilde{O}(k d \delta M + k^2 d N)$	$d < O(\mu^{-k} k^k)^{\ddagger}$

^αThey assume all spatial hashing values are distinct.

^βThey assume that the grid cell points of stride δ for each item are distinct.

^γThey assume *all* axes separate each item, i.e., global disjoint projection.

[†]This requires to satisfy $d = O(1)$ and $d > \kappa$ at the same time, which is infeasible.

[‡] $\mu := \frac{M(2\delta+1)}{2^u}$, which is < 1 for typical parameters, e.g., $M \leq 2^{16}$, $\delta \leq 30$, $u = 32$.

Spatial Hashing Spatial hashing has been a central building block of FPSI protocols over the Euclidean space [26,27,24,2,38,3,36,17,9]. This technique allows for filtering out non-matching items via hashed values, effectively reducing the number of candidates for closeness matching. Many recent protocols have been built upon this framework by employing various cryptographic techniques, such as functional secret sharing [26,27,24], additive homomorphic encryption [2,23], distributed point function [9], and VOLE [3]. While the protocol by Garimella et al. [26]—the first FPSI construction based on spatial hashing—was for ℓ_∞ metric, subsequent studies, including van Baarsen and Pu [2,3], Richardson et al. [38], and Piske et al. [36], extend it for general ℓ_p distances.

Despite their great success, spatial hashing-based constructions exhibit $\Omega(2^d)$ communication and/or computation complexity, which is inherited from the matching ambiguity of 2^d candidates. This complexity constrains their protocol to be feasible only when $d = O(1)$, though such a parameter regime suffices for some applications, e.g., taxi driver matching [29] or contact tracing [19]. In

addition, these protocols typically require that all set items are far apart by at least 2δ or more for the decision threshold δ —so-called “apart by 2δ ” assumption. Structural assumptions regarding spatial hashing have also been proposed, such as the disjoint hash [38,36] or disjoint mini-universe [24,9] assumptions, where spatial hashing values or grid cell coordinates do not overlap across set items.

Disjoint Projection Assumption To mitigate $\Omega(2^d)$ overhead, several works study additional structural assumptions stronger than “apart by $> 2\delta$ ”. Among them, the disjoint projection assumption showed the feasibility of designing an efficient FPSI with $O(\text{poly}(d))$ overhead [26,2,3,23]. The disjoint projection assumption requires that, for every item x , there exists an axis j such that no other item lies within distance δ of x along the j^{th} coordinate.

This assumption was first studied by Garimella et al. [26], who propose the first FPSI protocol for ℓ_∞ metric with $O(d^2)$ overhead. Note that the construction of Garimella et al. relies on a stronger form: for every item and axis, no other item lies within distance δ along the axis. Later, van Baarsen and Pu [2] relaxed their assumption to the disjoint projection assumption as stated above, maintaining the same $O(d^2)$ overhead. Gao et al. [23] further improved this complexity by $O(d)$, proposing other construction for ℓ_1 , ℓ_2 , and Hamming metrics.

The plausibility of the disjoint projection assumption is often justified for high-dimensional data. In particular, van Baarsen and Pu proved that, for the uniformly sampled data points, the disjoint projection assumption holds with a probability of $1 - \text{negl}(d)$ for some negligible function negl [2, Lemma 7]. Since the correctness of these protocols relies on the disjoint projection assumption, this result enforces $d > \kappa$ for the statistical security parameter κ [3].

Constructions from Other Approaches There exist FPSI proposals that do not rely on separation-based assumptions and spatial hashing [40,13,16,6,39,8]. They often directly model similarity matching (or its efficient approximation) as a circuit, and employ secure computation techniques for evaluating such a circuit, e.g., VOLE-based computation [13,8] or fully homomorphic encryption [40,39,8]. While these constructions do not require structural assumptions on the data in general, to the best of our knowledge, these constructions require $O(MN)$ computation and/or communication complexity overhead derived from pairwise similarity matching [40,13,6,39,8]. Although some constructions showed that a subquadratic complexity is achievable [16], they require another variant of separation-based assumption to avoid the quadratic blow-up. Recently, van Baarsen and Pu [3] proposed an FPSI protocol based on prefix tree representation without relying on separation-based assumptions. While their protocol achieves linear complexity in both M and N , it exhibits $O((\log \delta)^d)$ overhead.

2 Technical Overview

We briefly overview the high-level idea of the proposed FPSI protocol and present the core motivation of introducing the k -max collision assumption. Details will

be provided in the subsequent sections. Throughout the paper, we consider the receiver–sender model, where the receiver learns the fuzzy intersection result while the sender learns nothing after invoking the protocol.

2.1 FPSI with Matching Multiplicity

In prior FPSI with separation assumptions, each receiver’s ($\mathcal{R}$) item matches at most one of the sender’s ($\mathcal{S}$) items. However, when abandoning the separation-based paradigm, the functionality must capture multiplicity; $\mathcal{R}$ ’s item can match multiple $\mathcal{S}$ ’s items. To capture this, a straightforward approach is to simply output all the matched items, likewise to a few protocols without assumptions [6,3]. Instead, we take a different approach: model the δ -closeness as an oblivious incidence relationship instead of explicitly revealing matched items. Concretely, we represent this incidence relationship as a bipartite graph with an edge between $\mathcal{R}$ ’s item and $\mathcal{S}$ ’s items whenever they are δ -close, where $\mathcal{S}$ ’s indices are kept anonymous. Then $\mathcal{R}$ learns the incident structure without explicitly recovering the sender’s matched items.

2.2 Strawman Protocol and Its Limitation

As a warm-up, we consider the following strawman protocol, which will be the backbone of our construction, and discuss its limitations.

Testing Closeness from Membership Tags For the two parties, the receiver $\mathcal{R}$ and the sender $\mathcal{S}$, let us consider the following simple yet insecure protocol. We consider $\mathcal{U} = [0, 2^u) \cap \mathbb{Z}$ as the universe set, and denote $X, Y \subset \mathcal{U}^d$ such that $|X| = M$ and $|Y| = N$ as the sets held by $\mathcal{R}$ and $\mathcal{S}$, respectively. For each $x_i \in X$ and the index $j = 1, \dots, d$, we denote $x_{i,j}$ as the j^{th} component of x_i . We denote $y_{i,j}$ for each $y_i \in Y$ in a similar manner.

The first step of $\mathcal{R}$ is to encode the membership tag for each axis, which indicates the possible candidates of being matched. More precisely, for each axis j and the position $v \in \mathcal{U}$ we define

$$S_{v,j} := \{i \in \{1, \dots, M\} : v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}\}. \quad (1)$$

That is, $S_{v,j}$ encodes the candidates of matched item indices in X when the given item takes v at the j^{th} coordinate. In particular, if an element $y_i \in \mathcal{U}^d$ matches with some $x_{i'}$, i.e., $\|x_{i'} - y_i\|_\infty \leq \delta$, then for all $j = 1, \dots, d$, $i' \in S_{y_{i,j},j}$ holds. This is equivalent to $i' \in \bigcap_{j=1}^d S_{y_{i,j},j}$.

From this observation, $\mathcal{R}$ can encode X as the set $\mathcal{T}$ of tuples defined as follows, which has a cardinality of at most $(2\delta + 1)dM$.

$$\mathcal{T} := \left\{ (j, v, S_{v,j}) : j = 1, \dots, d \wedge v \in \bigcup_{i=1}^M ([x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}) \right\}. \quad (2)$$

Then $\mathcal{R}$ sends $\mathcal{T}$ to $\mathcal{S}$.

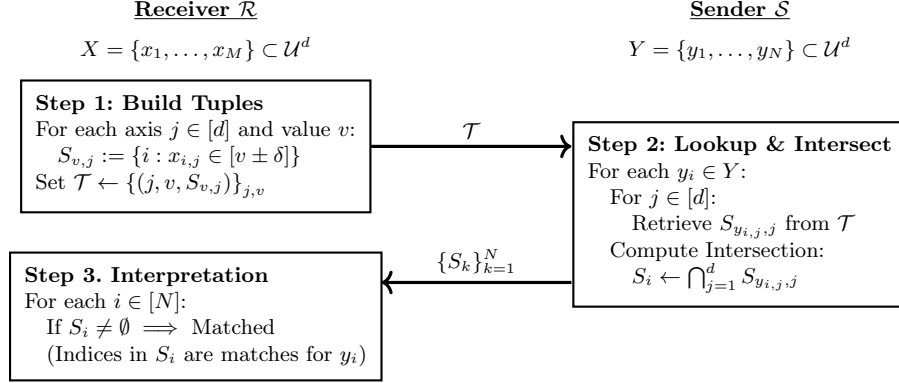


Fig. 2: Strawman protocol from membership tags

After receiving these tuples, for each $y_i \in Y$ and $j = 1, \dots, d$, $\mathcal{S}$ collects the $S_{y_{i,j},j}$; if there is no tuple that starts with $(j, y_{i,j})$, then set $S_{y_{i,j},j}$ as an empty set. For each $i = 1, \dots, N$, $\mathcal{S}$ computes $S_i \leftarrow \bigcap_{j=1}^d S_{y_{i,j},j}$, and send the collection $\{S_i\}_{i=1}^N$ to $\mathcal{R}$. Finally, $\mathcal{R}$ can determine the matched items by examining whether S_i is empty or not; if $i' \in S_j$ for some $i' \in \{1, \dots, M\}$, then $\mathcal{R}$ can conclude that $\|x_{i'} - y_i\|_\infty \leq \delta$. We illustrate the above procedures in Fig. 2.

Secure Realization to FPSI – A High-level Intuition To make this protocol secure, several standard cryptographic tools and techniques can be utilized. At this moment, we provide a high-level intuition only; we deliberately put subtle issues—e.g., serializing sets with supporting set intersection—aside and will present a concrete construction in the next subsections.

When sending the tuples in Eq. (2), $\mathcal{R}$ can employ the oblivious key-value store (OKVS) [25], with an appropriate serialization of $S_{v,j}$. OKVS encodes key-value pairs $\{(k_i, v_i)\}$ into some data structure OKVS, which supports decoding of k_j to v_j that was previously encoded; if not, then the algorithm returns a random string. Recent OKVS constructions feature encoding time and OKVS size linear in the number of key pairs, and support constant-time decoding for a single element [25,37,5]. In our context, for a cryptographic hash function H , $\mathcal{R}$ can encode the elements of $\mathcal{T}$ in Eq. (2) via $(H(v\|j), S_{v,j})$. Note that, to hide the cardinality of $\mathcal{T}$, $\mathcal{R}$ should additionally encode dummy key-value pairs to make the total number of encoded pairs in the OKVS exactly $(2\delta + 1)dM$, with a suitable domain separation on the key to prevent overlaps. In addition, oblivious computation of set intersection (i.e., $\bigcap_{j=1}^d S_{y_{i,j},j}$) has been studied for decades, and $\mathcal{S}$ can leverage rich tools and techniques [22,31,15,35,37]. In particular, we leverage a polynomial encoding of each set, so that the intersection can be computed via a random linear combination of the encodings [31].

Challenge: Blow-up on Tag Representations While this membership-tag-based approach appears to yield a plausible FPSI protocol, it is inefficient due to the size of the membership tag. To represent the membership of M elements,

the size of each (serialized) membership tag $S_{v,j}$ becomes $\Omega(M)$ as we need to assign 1-bit (included or not) information for each element. That is, although the number of key-value pairs to be encoded remains $O(M)$, the underlying data format of OKVS, which is typically a finite field element, should accommodate $\Omega(M)$ bits. In addition, $\mathcal{R}$ should retrieve N membership tags from $\mathcal{S}$ that contain the matching results, which incurs $\Omega(MN)$ overhead. For this reason, the strawman protocol could not avoid the $\Omega(MN)$ communication overhead.

2.3 Mitigating Tag Blow-up via k -Max Collision Assumption

Our key observation is that, if we can ensure that $|S_{v,j}|$ is always less than some fixed constant, say k , one can serialize $S_{v,j}$ as a polynomial $\prod_{z \in S_{v,j}} (x - z)$ of degree at most k , which is defined over a finite field $\mathbb{F}_p$ with $p \geq M$. That is, $O(k \log M)$ bits suffice for serialization. In particular, the $\log M$ term can be absorbed into the OKVS field size ($\geq 2^\kappa$), so $O(k)$ field elements are sufficient.

Motivated by this observation, we introduce a new structural assumption, called *k -max collision*, that enforces $|S_{v,j}|$ to be at most a parameter k . From the definition of $S_{v,j}$, this naturally corresponds to the maximum number of intersected items in terms of δ -neighborhoods for each axis. The formal description and its plausibility analysis are provided in the following paragraphs:

k -Max Collision Property Under the same notation as the previous section, let us consider a set $S \subset \mathcal{U}^d$. We say that S satisfies the k -max collision property of radius δ if the following inequality holds.

$$\max_{j=1,\dots,d} \left\{ \max_{v \in \mathcal{U}} |\{x_i \in S : v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}\}| \right\} \leq k.$$

Intuitively, along each axis, at most k items may overlap within any δ -neighborhood. Note that this directly imposes $|S_{v,j}| \leq k$ because of its definition. In addition, to avoid the trivial case, k should be strictly less than $|S|$.

Unlike separation-based assumptions, including “apart by 2δ ” and disjoint projection, this property permits arbitrary proximity between items, as long as the multiplicity of overlap along each coordinate remains bounded. For this reason, the k -max collision property is inherently incomparable with existing separation-based assumptions. We provide the following examples where the k -max collision property holds, while the others fail:

- (k -max $\nRightarrow$ “Apart by 2δ ”) Consider a multiset $S_1 = \{(0,0), (0,0), (3,3)\}$ for $\delta = 1$, i.e., two points are exactly overlap. Although there are exactly overlapping items, S_1 satisfies 2-max collision.
- (k -max $\nRightarrow$ disjoint projection) Consider $S_2 = \{(0,0), (0,3), (3,0)\}$ for $\delta = 1$. Although S_2 fails to satisfy disjoint projection, it satisfies 2-max collision.

Similarly, one can easily produce the opposite case where all set items are aligned with a specific axis, e.g., $S_3 := \{(0,-3), (0,0), (0,3)\}$ For $\delta = 1$, both “apart by 2δ ” and disjoint projection hold owing to the 2nd axis, while failing the k -max collision property for all $k < 3$. Fig. 3 visualizes all the incompatible cases S_1 , S_2 , and S_3 .

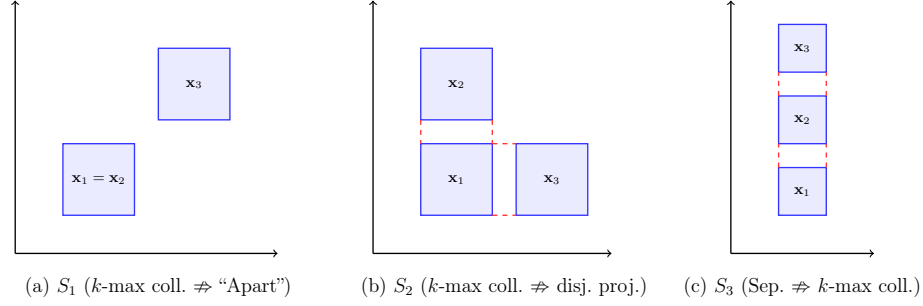


Fig. 3: Incompatible cases between k -max collision and separation assumptions. k -max collision holds in (a) and (b) but “apart by 2δ ” or disjoint projection fails. In contrast, both separation assumptions hold in (c) but k -max collision fails.

Plausibility of the k -Max Collision Assumption Despite its clear motivation, determining the plausible range of k is nontrivial. Since the parameter k restricts the density of the set items for each axis, the assumption would be more likely to hold when the data is evenly spread over the universe. In addition, the sparsity of δ -neighborhood interval $[x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}$ compared to the whole universe $\mathcal{U}$ also helps reducing the collision. With these insights, similar to disjoint projection assumptions [2,23,43], we prove that the k -max collision property holds with an overwhelming probability for the set of randomly sampled items over $\mathcal{U}^d$. The informal version is provided in Theorem 1, and full statement and proof will be given in Section 5.2.

Theorem 1 (Informal). For $A = \kappa + u + \log d$ and $\mu = \frac{M(2\delta+1)}{2^u}$ such that $A > \frac{e^2\mu}{\ln 2}$, it suffices to choose $k \geq \max \left\{ \mu e, \frac{A}{\log A - \log e - \log \mu + 1} \right\}$ to satisfy k -max collision assumption with a probability at least $1 - 2^{-\kappa}$ for a set that consists of M elements randomly sampled from $\mathcal{U}^d$.

If we fix κ, u, μ as constants, then $A = O(\log d)$ and k scales as $\frac{\log d}{\log \log d}$. Note that μ is substantially smaller than 1 in typical parameters, e.g., $\mu < 2^{-10}$ when $(M, u, \delta) = (2^{16}, 32, 30)$. Specifically, disjoint projection assumptions [26,2,43,23] implicitly require $\mu < 1$ to avoid the inevitable overlaps from the pigeonhole principle. In contrast, we can still select a valid k even when $\mu \geq 1$, provided that the specified conditions in Theorem 1 hold. These observations indicate the robustness of the k -max collision assumption with respect to d and μ . We provide a detailed scaling law with different parameter presets in Section 5.3.

Notably, the axis-wise density captured by the k -max collision property characterizes the resolution of each axis, i.e., the extent to which items can be distinguished from each other. When $k = 1$, every axis fully discriminates the $\mathcal{R}$ ’s items, whereas if k hits M due to some axis, that axis provides no discriminative power. Thus, k quantifies the worst-case ambiguity when projected to a single axis. In addition, the k -max collision property is locally verifiable in time $O(udM)$ via Algorithm 1 in Appendix B, ensuring that the assumption can be validated in advance rather than remaining purely hypothetical.

2.4 FPSI with k -max Collision Assumption

With the k -max collision assumption, we can construct the FPSI protocol from the membership tag approach. We provide details on a secure realization of the protocol in Section 2.2 through the polynomial encodings of the sets. To compute intersections over these encodings, we adopt the technique by Kissner and Song [31]. The resulting protocol achieves $O(kd\delta M + kN)$ communication and $O(k \log^2 k \cdot (d\delta M + N) + k^2 dN)$ computation complexity.

Membership Tags and Their Intersection From k -max collision assumption, we know that $|S_{v,j}| \leq k$. Hence, for a finite field $\mathbb{F}$, $\mathcal{R}$ can encode $S_{v,j}$ as $f_{v,j}(x) := \prod_{z \in S_{v,j}} (x - z) \cdot \zeta_{v,j}(x) \in \mathbb{F}[x]$ of degree at most k , where $\zeta_{v,j}(x)$ is a random polynomial of degree at most $k - |S_{v,j}|$. The intersection $\bigcap_{j=1}^d S_{y_i,j}$ then corresponds to the zeros of the random linear combination of $f_{y_i,j,j}(x)$, namely, $g_{y_i}(x) := \sum_{j=1}^d \gamma_{i,j}(x) f_{y_i,j,j}(x)$ for $\gamma_{i,j}(x)$ of degree $\leq k - 1$ whose coefficients are randomly sampled from $\mathbb{F}$. For $S_{y_i} := \bigcap_{j=1}^d S_{y_i,j}$, the distributions $g_{y_i}(x)$ and $\prod_{z \in S_{y_i}} (x - z) \cdot \zeta^*(x)$ have a statistical distance at most $O(\frac{kd}{|\mathbb{F}|})$, where $\zeta^*(x)$ is a random polynomial of degree at most $2k - |S_{y_i}| - 1$ (See Lemma 5 for details). In addition, if $z \in S_{y_i}$, then $g_{y_i}(z) = 0$, and for $z \notin S_{y_i}$, $g_{y_i}(z)$ follows the uniform distribution over $\mathbb{F}$, i.e., $\Pr[g_{y_i}(z) = 0] = \frac{1}{|\mathbb{F}|}$.

By representing each polynomial in terms of its coefficients, $g_{y_i}(x)$ can be obviously computed with an AHE whose message space is $\mathbb{F}$. More precisely, we consider the “ $\star$ ” operation defined as below:

$$(a_1, \dots, a_n) \star (b_1, \dots, b_m) := \sum_{j=1}^n a_j \odot (\underbrace{0, \dots, 0}_{j-1}, b_1, \dots, b_m, \underbrace{0, \dots, 0}_{n-j}) \in \mathbb{F}^{n+m-1}.$$

Note that we can compute for encrypted $b_1, \dots, b_m$ and scalars $a_1, \dots, a_n$. The following equality holds for a vector $\mathbf{x} := (1, x, \dots, x^{n+m-1})$:

$$\langle (a_1, \dots, a_n) \star (b_1, \dots, b_m), \mathbf{x} \rangle = \left(\sum_{i=1}^n a_i x^{i-1} \right) \cdot \left(\sum_{j=1}^m b_j x^{j-1} \right).$$

That is, if we regard the coefficients of $\gamma_{i,j}(x)$ as a_i ’s and those of $f_{y_i,j,j}(x)$ as b_i ’s, then $g_{y_i}(x)$ can be represented as the sum of star products with indices $j = 1, \dots, d$.

From this insight, $\mathcal{R}$ encodes each polynomial $f_{v,j}(x)$ via a $(k+1)$ -dimensional vector $(f_{v,j,1}, f_{v,j,2}, \dots, f_{v,j,k+1}) \in \mathbb{F}^{k+1}$ such that $f_{v,j}(x) = \sum_{m=1}^{k+1} f_{v,j,m} x^{m-1}$; we set $f_{v,j,m} = 0$ if $m \geq \deg(f_{v,j})$. Then under the same notation with Eq. (2), $\mathcal{R}$ make an OKVS with the key-value pairs

$$(\mathbf{H}(v||j), \llbracket f_{v,j,1} \rrbracket \parallel \llbracket f_{v,j,2} \rrbracket \parallel \dots \parallel \llbracket f_{v,j,k+1} \rrbracket),$$

where $\llbracket x \rrbracket$ denotes the encryption of x by an AHE with $\mathcal{R}$ ’s key. Each AHE ciphertext should be encrypted with different randomness. Note that, as mentioned

earlier, $\mathcal{R}$ additionally encodes dummy key-value pairs to make the number of encoded entries exactly $(2\delta + 1)dM$.

Upon receiving the OKVS, for each $\mathcal{S}$'s item $y_i \in Y$, $i = 1, \dots, N$, and $j = 1, \dots, d$, $\mathcal{S}$ can retrieve the ciphertexts corresponding to $f_{y_i, j}(x)$ and compute the “ $\star$ ” operation over the encrypted domain with a uniformly sampled $\gamma_{i, j, n}$ from $\mathbb{F}$ for $n = 1, \dots, k$:

$$\mathbf{S}_i = \sum_{j=1}^d \sum_{n=1}^k \gamma_{i, j, n} \odot \underbrace{(0, \dots, 0, \llbracket f_{y_i, j, 1} \rrbracket, \dots, \llbracket f_{y_i, j, k+1} \rrbracket)}_{n-1}, \underbrace{(0, \dots, 0)}_{k-n}.$$

If $H(y_{i, j} \| j)$ was not encoded in the received OKVS, then the decoded value becomes a random string of the same length as the ciphertext. Note that each $\mathbf{S}_i$ consists of $2k$ ciphertexts. Finally, $\mathcal{S}$ sends $\{\mathbf{S}_i\}_{i=1}^N$ to the receiver, while shuffling the index with respect to i .

After receiving $\{\mathbf{S}_i\}_{i=1}^N$ from $\mathcal{S}$, $\mathcal{R}$ can recover the plaintext polynomials $\{\hat{f}_i(x)\}_{i=1}^N$ by decrypting all the ciphertexts and regard them as coefficient. Since we know $\deg(\hat{f}_i(x)) \leq 2k - 1$, $\mathcal{R}$ can factorize the retrieved $\hat{f}(x)$ through classical algorithms with NTT-based polynomial multiplication [11, 41]. In particular, since all we need is its linear factor, one can utilize the following simpler algorithm when $\mathbb{F}$ is an NTT-friendly prime field, which takes $O(k \log^2 k \log |\mathbb{F}|)$ field operations [41, Corollary 14.16].

If $\hat{f}_i(x)$ has a zero $v \in \{1, \dots, M\}$, then $\mathcal{R}$ can ensure that x_v is matched with some $\mathcal{S}$'s item with high probability; for each $\mathcal{S}$'s item, the false positive probability can be bounded by $\frac{M}{|\mathbb{F}|}$, which is obtained by applying the union bound on $\frac{1}{|\mathbb{F}|}$ over at most M candidates of wrongly matched indices.

Complexity Analysis We briefly describe the complexity analysis to demonstrate that the proposed approach retains linear complexity in both the set sizes (M, N) and d , under the bounded parameter k . For simplicity, we omit the constant factors induced by the computational and statistical security parameters. The full complexity analysis will be given in Section 6.2.

- (Communication) The number of encoded key-value pairs in the OKVS is at most $(2\delta + 1)dM$, and each value has a size of $(k + 1) \cdot |\text{ct}|$, where $|\text{ct}|$ denotes the size of an AHE ciphertext. On the other hand, $\mathcal{S}$ sends $\{\mathbf{S}_i\}_{i=1}^N$ to $\mathcal{R}$, which consists of $2kN$ AHE ciphertexts in total. Therefore, the total communication becomes $O(k\delta dM + kN)$.
- ($\mathcal{R}$'s Computation) To construct the key-value pairs of the OKVS, the receiver needs to compute at most $(2\delta + 1)dM$ polynomials $f_{v, j}(x)$, and computing each of them takes $O(k \log^2 k)$ field operations due to its structure. After recovering $\hat{f}_i(x)$ from $2kN$ decryptions, their factorization over $\mathbb{F}$ takes $O(k \log^2 k \log |\mathbb{F}|)$ field operations [11, 41]. Thus, when regarding $|\mathbb{F}|$ as a constant, $\mathcal{R}$'s total computation becomes $O(k \log^2 k \cdot (\delta dM + N))$.
- ($\mathcal{S}$'s Computation) $\mathcal{S}$ computes dN OKVS decodings and $k^2 dN$ operations over the AHE ciphertexts. Since OKVS supports constant-time decoding, the total computation of $\mathcal{S}$ becomes $O(k^2 dN)$.

Parameters: Two parties, the receiver $\mathcal{R}$ and the sender $\mathcal{S}$. The universe set $\mathcal{U}$ and the data dimensionality $d \in \mathbb{N}$. The threshold $\delta > 0$. Inputs: – $\mathcal{R}$: A set $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$ of cardinality M. – $\mathcal{S}$: A set $Y = \{y_1, \dots, y_N\} \subset \mathcal{U}^d$ of cardinality N. Outputs: Upon receiving X, Y from $\mathcal{R}$ and $\mathcal{S}$, the functionality $\mathcal{F}_{\text{FPSI}}$ randomly samples a permutation $\pi : [N] \rightarrow [N]$ and returns – $\mathcal{R}$: A collection of sets $\{S_i \subset [N]\}_{i \in [M]}$ such that $j \in S_i$ iff $\ x_i - y_{\pi(j)}\ _\infty \leq \delta$. – $\mathcal{S}$: Nothing.
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Fig. 4: The ideal functionality $\mathcal{F}_{\text{FPSI}}$ for ℓ_∞ metric.

3 Preliminaries

3.1 Notations

For $d \in \mathbb{N}$, we denote $[d]$ as the set $\{1, \dots, d\}$. For a finite set S , we denote $x \xleftarrow{\$} S$ as a uniformly random sample over S . For a field $\mathbb{F}$, we denote $\mathbb{F}[X]^{\leq k}$ as the set of polynomials defined over $\mathbb{F}$ of degree at most k . For probability distributions $\mathcal{A}$ and $\mathcal{B}$, the notation $\mathcal{A} \approx_c \mathcal{B}$ means that they are computationally indistinguishable. Throughout this paper, we denote λ and κ as computational and statistical security parameters, respectively. We denote $\mathcal{U}$ as the universe of u -bit unsigned integers for $u \in \mathbb{N}$, i.e., $\mathcal{U} = [0, 2^u) \cap \mathbb{Z}$. Unless specified, we use $\log$ as the logarithm of base 2 and denote $\ln$ as the natural logarithm. We denote $\|\cdot\|_\infty$ as the ℓ_∞ norm, i.e., for $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{Z}^d$, $\|\mathbf{x}\|_\infty := \max_{i \in [d]} \{ |x_i| \}$.

3.2 Fuzzy Private Set Intersection

Ideal Functionality with Matching Multiplicity Throughout this paper, we consider the receiver ($\mathcal{R}$) and the sender ($\mathcal{S}$) scenario where the receiver only retrieves the protocol output. Unlike many existing protocols from separation-based paradigms, in our setting, each $\mathcal{R}$'s item could be matched with multiple items of $\mathcal{S}$. This is the consequence of abandoning separation-based assumptions. Note that this situation also appears in recent fuzzy PSI constructions without structural assumptions [24, 3]. They simply permit $\mathcal{R}$ to learn all the matched items by $\mathcal{S}$, likewise to the usual definitions of FPSI functionality.

Instead of revealing $\mathcal{S}$'s matched items, we consider the functionality where $\mathcal{R}$ learns an oblivious incidence relationship induced by δ -closeness. This incidence relationship is represented as a bipartite graph $G = (V, E)$ where $V = A \cup B$ for $A = [M]$ and $B = [N]$, where B is an anonymous index set corresponding to $\mathcal{S}$'s items. $(i, j) \in E$ if and only if $\|x_i - y_j\|_\infty \leq \delta$, where the indices in B are randomly permuted. $\mathcal{R}$ learns (A, E) , while the indices B remain hidden. This functionality formalizes δ -closeness as a relation, rather than as an explicit recovery of $\mathcal{S}$'s matched items. Fig. 4 describes the ideal functionality $\mathcal{F}_{\text{FPSI}}$.

Semi-Honest Adversary Model For a two-party functionality $\mathcal{F}$ between parties P_1, P_2 , we say that a protocol Π securely realizes $\mathcal{F}$ under the semi-honest model if there is a simulator Sim that takes the input and output of $\mathcal{F}$, producing the view View of each party during the protocol, including protocol input, internal randomness, and the transcript. Formally, for $i \in \{1, 2\}$ the following two distributions must be computationally indistinguishable:

$$\{\text{Sim}(i, x_i, f(x_1, x_2), 1^\lambda)\} \approx_c \{\text{View}(\{\Pi, P_i, x_1, x_2, 1^\lambda\})\}$$

3.3 Oblivious Key-Value Store

Oblivious key-value store (OKVS) [25] is a data structure for key-value pairs that supports oblivious retrieval. That is, it decodes valid values for previously encoded keys; otherwise, it returns a random value. Several (F)PSI protocol constructions adopted it as a core building block [26, 2, 23, 37, 5].

Definition 1 (Oblivious Key-Value Store [25]). Let $\mathcal{K}, \mathcal{V}$ be key and value spaces, respectively. An oblivious key-value store (OKVS) defined over $(\mathcal{K}, \mathcal{V})$ is a data structure that consists of two algorithms $(\text{Encode}, \text{Decode})$ defined by

- **Encode:** It is a probabilistic algorithm that takes a set of key-value pairs $\{(k_i, v_i)\} \subset \mathcal{K} \times \mathcal{V}$, returning an OKVS object S or a failure indicator $\perp$.
- **Decode:** It is a probabilistic algorithm that takes an OKVS object S and a key $k \in \mathcal{K}$, returning a value $v \in \mathcal{V}$.

OKVS must satisfy the following properties:

- (**Correctness**) For all $L = \{(k_i, v_i)\} \subset \mathcal{K} \times \mathcal{V}$ with distinct keys, if a key-value pair $(k_i, v_i) \in L$ and $\text{Encode}(L; \gamma) \rightarrow S \neq \perp$, then $\text{Decode}(S, k; \gamma) = v$.
- (**Obliviousness**) For any $\{k_1^0, \dots, k_n^0\}$ and $\{k_1^1, \dots, k_n^1\}$ of distinct n keys, if Encode does not return $\perp$, then for any $\{v_1, \dots, v_n\} \xleftarrow{\$} \mathcal{V}$, the following two distributions are computationally indistinguishable:

$$\{S_0 \leftarrow \text{Encode}(\{(k_i^0, v_i)\}_{i \in [n]})\} \approx_c \{S_1 \leftarrow \text{Encode}(\{(k_i^1, v_i)\}_{i \in [n]})\}.$$

In addition to these requirements, likewise prior FPSI studies, we also require that the OKVS must satisfy the following *independence* property. This property is proven to be true in existing OKVS constructions [26, 5, 2].

Definition 2 (Independence [25]). Let $\text{OKVS} = (\text{Encode}, \text{Decode})$ be an OKVS. We say that OKVS satisfies the independence property if for all key-value pairs $L = \{(k_i, v_i)\} \subset \mathcal{K} \times \mathcal{V}$ and a key k^* that do not appear in L , $\text{Decode}(\text{Encode}(L), k^*)$ is computationally indistinguishable from random in $\mathcal{V}$.

Existing OKVSs support a finite field $\mathcal{V}$, and when n key-value pairs are encoded, S is an element of $\mathcal{V}^m$ for $m = (1 + \epsilon)n$ with $\epsilon > 0$. Such an ϵ characterizes the efficiency of the OKVS, and the state-of-the-art construction exhibits $\epsilon = 0.03\text{--}0.1$ [5]. In addition, recent constructions achieve $O(\kappa n)$ and $O(\kappa)$ for OKVS encoding and decoding of n key-value pairs, respectively [25, 37, 5]. That is, the decoding time does not depend on the number of encoded key-value pairs.

3.4 Additive Homomorphic Encryption

We consider a symmetric-key version of an additive homomorphic encryption (AHE). For the key, message, and ciphertext spaces $(\mathcal{SK}, \mathcal{M}, \mathcal{C})$, respectively, we define an AHE as a tuple of algorithms $(\text{Setup}, \text{Enc}, \text{Add}, \text{SMul}, \text{Dec})$ as follows:

- $\text{Setup}(1^\lambda) \rightarrow \text{sk} \in \mathcal{SK}$: takes a security parameter and returns a secret key $\text{sk} \in \mathcal{SK}$.
- $\text{Enc}(\text{sk}, m; \gamma) \rightarrow \text{ct} \in \mathcal{C}$: takes a secret key sk , a plaintext $m \in \mathcal{M}$ and a randomness $\gamma \in \{0, 1\}^\lambda$, returning a ciphertext $\text{ct} \in \mathcal{C}$.
- $\text{Add}(\text{ct}_1, \text{ct}_2) \rightarrow \text{ct}^* \in \mathcal{C}$: takes two ciphertexts $\text{ct}_1, \text{ct}_2 \in \mathcal{C}$, which encrypts $m_1, m_2 \in \mathcal{M}$, resp., returning a ciphertext ct^* that decrypts to $m_1 + m_2$.
- $\text{SMul}(\text{ct}, c) \rightarrow \text{ct}^* \in \mathcal{C}$: takes a ciphertext $\text{ct} \in \mathcal{C}$ that encrypts $m \in \mathcal{M}$ and a scalar $c \in \mathcal{M}$, returning a ciphertext ct^* that decrypts to $c \cdot m$.
- $\text{Dec}(\text{sk}, \text{ct}) \rightarrow m \in \mathcal{M}$: takes a secret key sk and a ciphertext $\text{ct} \in \mathcal{C}$, returning a plaintext $m \in \mathcal{M}$.

For correctness, we assume that $\Pr[\text{Dec}(\text{sk}, \text{Enc}(\text{sk}, m)) = m] \geq 1 - \text{negl}(\kappa)$ holds for any $m \in \mathcal{M}$ and $\text{sk} \in \mathcal{SK}$. We denote $\oplus$ and $\odot$ as addition and scalar multiplication over an encrypted domain. For a circuit $\mathcal{C} : \mathcal{M}^k \times \mathcal{M}^l \rightarrow \mathcal{M}$ comprised by additions and scalar multiplications, $\text{Eval}(\mathcal{C}, (\text{ct}_1, \dots, \text{ct}_k), (y_1, \dots, y_l)) \rightarrow \text{ct}^*$ denotes the homomorphic circuit evaluation with plaintext scalars $(y_1, \dots, y_l)$.

Ciphertext Simulatability Our protocol design requires additional properties of AHE beyond standard IND-CPA security. We introduce ciphertext simulatability [30], which ensures that the AHE ciphertext can be simulated from the input plaintexts and the decryption result even after circuit evaluation. Since the original definition is for fully homomorphic encryptions, we adapt it as in Definition 3 for AHEs by restricting the type of supported circuit.

Definition 3 (Ciphertext Simulatability for AHE; Adapted from [30, Definition 7]). *Let AHE be an AHE defined over $(\mathcal{SK}, \mathcal{M}, \mathcal{C})$. We say that AHE is ciphertext-simulatable for a circuit $\mathcal{C} : \mathcal{M}^k \times \mathcal{M}^l \rightarrow \mathcal{M}$ whose total degree with respect to first k variables is 1, if there exists a simulator Sim such that for any $\mathbf{x} := (x_1, \dots, x_k)$ and $\mathbf{y} := (y_1, \dots, y_l)$, the following distributions are computationally indistinguishable:*

$$\begin{aligned} & \{(\text{sk}, (\text{ct}_1, \dots, \text{ct}_k), \text{ct}^*) : \text{ct}^* \leftarrow \text{Eval}(\mathcal{C}, (\text{ct}_1, \dots, \text{ct}_k), (y_1, \dots, y_l))\}, \\ & \{(\text{sk}, (\text{ct}_1, \dots, \text{ct}_k), \text{ct}^*) : z \leftarrow \mathcal{C}(\mathbf{x}, \mathbf{y}); \text{ct}^* \leftarrow \text{Sim}(\text{sk}, (\text{ct}_1, \dots, \text{ct}_k), z)\}, \end{aligned}$$

where $\text{sk} \leftarrow \text{Setup}(1^\lambda)$ and $\text{ct}_i \leftarrow \text{Enc}(\text{sk}, x_i; \gamma_i)$ for $\gamma_i \leftarrow \{0, 1\}^\lambda$ and $i \in [k]$.

This property is necessary in our protocol design because the computation of ciphertexts by $\mathcal{S}$ involves $\mathcal{S}$'s random blind values. To ensure simulation-based security, we need to ensure that the resulting ciphertexts are simulatable without seeing such blind values. Remark that many existing AHEs defined over cyclic groups trivially satisfy this property [20, 34, 12], as the ephemeral randomness of the output ciphertext also follows the uniform distribution. On the other

hand, AHEs based on (R)LWE [21,7,28] require additional techniques to handle the output ciphertext's LWE noise, e.g., through noise flooding [1] or dynamic randomization during circuit evaluation [30].

OKVS Compatibility Recall that $\mathcal{R}$ encodes AHE ciphertexts to the OKVS and $\mathcal{S}$ later decodes them. It should be ensured that the correctly decoded AHE ciphertexts are indistinguishable from decoding failures; otherwise, $\mathcal{S}$ can infer whether each item is correctly decoded or not. This phenomenon arises from the semantic mismatch between the ciphertext space and the OKVS value space.

We resolve this mismatch by introducing a new property called *the OKVS compatibility*, which formalizes the necessary interface to connect the two spaces discussed above. This property requires the conversion algorithm between $\mathcal{C}$ and $\mathcal{V}^l$ for some parameter l called the expansion rate, ensuring that the converted $\text{ct} \in \mathcal{C}$ is indistinguishable from $\mathcal{V}^l$. Here, l means that the ciphertext can be split and encoded to different l OKVSs under the same corresponding key.

Definition 4 (OKVS Compatibility). *Let AHE be an AHE defined over $(\mathcal{SK}, \mathcal{M}, \mathcal{C})$ and OKVS be an OKVS defined over $(\mathcal{K}, \mathcal{V})$. For a fixed constant $l \in \mathbb{N}$, we say that the AHE is κ -compatible with the OKVS with an expansion rate l if there exist two efficient algorithms $\text{C2V} : \mathcal{C} \rightarrow \mathcal{V}^l$ and $\text{V2C} : \mathcal{V}^l \rightarrow \mathcal{C}$ satisfying the following properties:*

- (*Correctness*) For all $\text{ct} \in \mathcal{C}$, $\Pr[\text{V2C}(\text{C2V}(\text{ct})) = \text{ct}] \geq 1 - \text{negl}(\kappa)$ holds.
- (*Obliviousness*) For any plaintext $m \in \mathcal{M}$ and $\text{sk} \leftarrow \text{AHE.Setup}(1^\lambda)$, the following two distributions are computationally indistinguishable:

$$\{\mathbf{v} \leftarrow \text{C2V}(\text{ct}) : \gamma \xleftarrow{\$} \{0, 1\}^\lambda; \text{ct} \leftarrow \text{AHE.Enc}(\text{sk}, m; \gamma)\} \approx_c \{\mathbf{v} \xleftarrow{\$} \mathcal{V}^l\}$$

Note that several prior studies have implicitly utilized this property via a careful choice of AHE and OKVS. For example, van Baarsen and Pu [2], and Gao et al. [23] slightly modify the OKVS to handle the value space $\mathcal{V}$ as a cyclic group of prime order, which came up with the observation that recent OKVS constructions define the decoding as an inner product of a binary vector into the OKVS object in $\mathcal{V}^m$ [37,5]. On the other hand, Vos et al. [42] employed Elligator2 map [4] to make an ElGamal encryption over the elliptic curve group compatible with an OKVS, with a careful treatment for failure cases. Since the ElGamal encryption satisfies IND\$-CPA security [10], i.e., for any $\text{sk} \leftarrow \text{AHE.Setup}(1^\lambda)$ and $m \in \mathcal{M}$, $\{c \leftarrow \text{AHE.Enc}(\text{sk}, m; \gamma) : \gamma \xleftarrow{\$} \{0, 1\}^\lambda\} \approx_c \{c \xleftarrow{\$} \mathcal{C}\}$, these modifications make the ElGamal encryption OKVS compatible.

3.5 Useful Mathematical Lemmas

To show the validity of the proposed k -max collision assumption, we will use the multiplicative Chernoff bound over the binomial distribution (See, e.g., [33]).

Lemma 1 (Chernoff Bound for Binomial Distribution). *Let X be a random variable that follows a binomial distribution $\text{Bin}(n, p)$. We denote $\mu = np$. Then for $k = (1 + \xi)\mu$ with $\xi > 0$, the following inequality holds:*

$$\Pr[X > k] < \left(\frac{e^\xi}{(1 + \xi)^{1+\xi}} \right)^\mu.$$

In particular, if $k > e\mu$, then the above bound can be simplified by

$$\Pr[X > k] < \left(\frac{e\mu}{k} \right)^k.$$

4 Fuzzy Private Set Intersection with Membership Tags

In this section, we present the membership tag-based FPSI framework for ℓ_∞ . We then identify the key bottleneck regarding the size of membership tags, which can be effectively resolved with the k -max collision property.

4.1 Membership Encodings of Sets

We reformulate the FPSI problem over ℓ_∞ metric as an oblivious intersection over coordinate-wise membership encodings. Let $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$ be a set. We denote $x_i = (x_{i,1}, \dots, x_{i,d}) \in \mathcal{U}^d$ for $i \in [M]$. For each axis $j \in [d]$, define a set-valued function

$$M_j : \mathcal{U} \rightarrow \mathcal{P}([M]) \text{ such that } M_j(v) = \{i \in [M] : v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}\},$$

where $\mathcal{P}([M])$ denotes the collection of all subsets of $[M]$. We call the collection $\{M_j\}_{j=1}^d$ the *membership encoding* of the set X of radius δ . In addition, we refer to the evaluation results of membership encodings, which are subsets of $[M]$, as *membership tags*. From the definition, we can observe the following lemma holds:

Lemma 2. *Let $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$ be a set and a collection $\{M_j\}_{j=1}^d$ be the membership encoding of X with radius δ . For $y_k = (y_{k,1}, \dots, y_{k,d}) \in \mathcal{U}^d$ and $i \in [M]$, we have that $\|x_i - y_k\|_\infty \leq \delta \iff i \in \bigcap_{j=1}^d M_j(y_{k,j})$.*

Proof. By definition, $\|x_i - y_k\|_\infty \leq \delta \iff \forall j \in [d], |x_{i,j} - y_{k,j}| \leq \delta$, i.e., $y_{k,j} \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}$ and $i \in M_j(y_{k,j})$. This completes the proof. $\square$

The above lemma shows that fuzzy matching of y_k over the set X reduces to testing the non-emptiness of an intersection of d coordinate-wise membership encodings. Accordingly, we define the following functionality $\mathcal{F}_{\text{FPSI-ME}}$ in Fig. 5. Thanks to Lemma 2, securely realizing this functionality suffices to realize $\mathcal{F}_{\text{FPSI}}$. For this reason, our protocol design therefore focuses on an efficient and secure realization of $\mathcal{F}_{\text{FPSI-ME}}$.

Parameters: Two parties, the receiver $\mathcal{R}$ and the sender $\mathcal{S}$. The universe set $\mathcal{U}$ and the data dimensionality $d \in \mathbb{N}$. The threshold $\delta > 0$.

Inputs:

- $\mathcal{R}$: A set $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$ of cardinality M .
- $\mathcal{S}$: A set $Y = \{y_1, \dots, y_N\} \subset \mathcal{U}^d$ of cardinality N .

Outputs:

Upon receiving X, Y from $\mathcal{R}$ and $\mathcal{S}$, the functionality $\mathcal{F}_{\text{FPSI}}$ randomly samples a permutation $\pi : [N] \rightarrow [N]$ and returns

- $\mathcal{R}$: A set $\{\bigcap_{j=1}^d M_j(y_{\pi(i),j}) : y_i = (y_{i,j})_{j=1}^d \in Y\}$, where $\{M_j\}_{j=1}^d$ denotes the membership encoding of X of radius δ .
- $\mathcal{S}$: Nothing.

Fig. 5: The ideal functionality $\mathcal{F}_{\text{FPSI-ME}}$.

Contrast with Prior Separation-based Paradigms In separation-based protocol designs, the matching process with δ -closeness is typically enforced via geometric filtering approaches such as spatial hashing or leveraging the separating axis guaranteed by the disjoint projection assumption. In contrast, our formulation decouples δ -closeness as a coordinate-wise membership relation captured by the membership encodings, without requiring geometric separation. This abstraction shifts the focus from geometric separation to structural properties of coordinate-wise overlap.

4.2 Protocol Design Insight and Key Bottleneck

We present our insight to design an FPSI protocol based on realizing $\mathcal{F}_{\text{FPSI-ME}}$. Throughout this section, we only provide a high-level intuition; the complete protocol construction and detailed analyses are provided in Section 6.

Serialization of Membership Tags To realize $\mathcal{F}_{\text{FPSI-ME}}$, each output set from the membership encoding should be carefully serialized. The serialization and its inverse should be efficiently done, and the set intersection operation can also be done in a serialized form. In addition, the intersection after serialization should not provide more information than the actual set intersection result. Formally, for a finite field $|\mathbb{F}|$, constants $l \in \mathbb{N}$, $d \in \mathbb{N}$, and the statistical security parameter κ , we consider probabilistic algorithms $\text{Ser} : \mathcal{P}([M]) \rightarrow \mathbb{F}^l$, $\text{DeSer} : \mathbb{F}^l \rightarrow \mathcal{P}([M])$, and $\text{Inter} : \prod_{i=1}^d \mathbb{F}^l \rightarrow \mathbb{F}^l$ with the following properties:

- (**Serialization Correctness**) For any $S \in \mathcal{P}([M])$,

$$\Pr[\text{DeSer}(\text{Ser}(S)) = S] \geq 1 - \text{negl}(\kappa).$$

- (**Intersection Correctness**) For any $S_1, \dots, S_d \in \mathcal{P}([M])$,

$$\Pr \left[\text{DeSer} \left(\text{Inter}(\text{Ser}(S_1), \dots, \text{Ser}(S_d)) \right) = \bigcap_{i=1}^d S_i \right] \geq 1 - \text{negl}(\kappa),$$

- (**Intersection Obliviousness**) For sets $S_1, \dots, S_d$ and $S := \bigcap_{i=1}^d S_i$, there exists a simulator Sim s.t. the following two distributions have statistical distance at most $\text{negl}(\kappa)$: $\{\mathbf{f} \leftarrow \text{Inter}(\text{Ser}(S_1), \dots, \text{Ser}(S_d))\}$ and $\{\mathbf{f} \leftarrow \text{Sim}(S)\}$.

By a slight abuse of notation, we also regard the serializations of sets as membership tags. In particular, we say that the membership tag is *algebraic* if Inter can be implemented as an algebraic circuit over $\mathbb{F}$.

With this serialization, we can represent the membership encoding $\{M_j\}_{j=1}^d$ of X by applying Ser for each membership tag. Precisely, we re-write each M_j as the mapping from $\mathcal{U}$ to $\mathbb{F}^l$ such that

$$v \mapsto \text{Ser}(\{i \in [M] : v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}\}).$$

If the serialization is algebraic, we can represent the evaluation of $\bigcap_{j=1}^d M_j(y_{i,j})$ in a serialized form as an arithmetic circuit.

Viewing $\mathcal{F}_{\text{FPSI-ME}}$ as a Secure Computation The membership tag-based abstraction indicates that we can realize $\mathcal{F}_{\text{FPSI-ME}}$ as a secure computation of (i) looking up each membership encoding and (ii) evaluating Inter . In particular, if the serialization is algebraic, then the latter process can be represented as an arithmetic circuit. From this viewpoint, several realizations may be possible, such as modeling the whole process via an arithmetic circuit and leveraging a generic two-party computation framework. In Section 6.1, we will present our realization through (i) an OKVS and (ii) a ciphertext simulatable and κ -compatible AHE, and (iii) polynomial-based set serialization inspired by Kissner and Song [31].

Key Bottleneck: Quadratic Overhead from Tag size While this approach provides a new abstraction for realizing $\mathcal{F}_{\text{FPSI}}$, a key bottleneck remains: the size of the membership tag grows in the $\mathcal{R}$'s set size M . More precisely, in the absence of structural constraints, any injective encoding of subsets of $[M]$ requires $\Omega(M)$ bits. Since $\mathcal{F}_{\text{FPSI-ME}}$ outputs a collection of N membership tags, the overall output size is $\Omega(MN)$ bits. Therefore, any protocol realizing $\mathcal{F}_{\text{FPSI-ME}}$ must communicate at least $\Omega(MN)$ bits, i.e., quadratic overhead in communication is unavoidable. This motivates controlling the matching multiplicity via structural constraints, so that each membership tag admits a compressed representation, thereby reducing the overall communication below the $\Omega(MN)$ regime.

5 k -Max Collision Property

We resolve the aforementioned blow-up issue on the tag size by introducing a new structural assumption called the *k-max collision*, which explicitly controls the matching multiplicity for each axis. Following the setting in disjoint projection assumptions [2,23,43], we investigate its feasibility for a randomly sampled element over the universe. Unlike disjoint projection assumptions [2,23,43], this new density-bounded assumption does not require the data dimension to be high enough, e.g., $d > \kappa$ for the statistical security parameter κ .

With the membership tag-based abstraction in Section 4.2, we construct the proposed FPSI protocol under the k -max collision assumption, i.e., restricting the size of each membership tag to at most k . The detailed construction will be presented in Section 6.1.

5.1 Motivation and Definition

Our key observation is that the lower bound $\Omega(M)$ of the tag size stems from the worst-case configuration where δ -neighborhoods of every item overlap when projected into some axis. In contrast, when the axis projections of the items are not densely clustered, the overlaps along each axis are typically much smaller. This suggests that the $\Omega(M)$ tag size is not inherent to our membership tag-based formulation itself, but the density of the items projected onto each axis. Motivated by this structural insight, we introduce the *k -max collision property*, which explicitly bounds the maximum number of overlaps of δ -neighborhoods by a density parameter k .

Definition 5 (k -Max Collision). Let $\mathcal{U} := [0, 2^u) \cap \mathbb{Z}$ be the universe set. For parameters $k, \delta \in \mathbb{N}$ and a set $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$, we say that X has a k -max collision property of radius δ if the following inequality holds:

$$\max_{j \in [d]} \left\{ \max_{v \in \mathcal{U}} |\{x_i \in X : v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}\}| \right\} \leq k,$$

where $x_{i,j}$ denotes the j^{th} coordinate of $x_i \in \mathcal{U}^d$.

We remark that this property trivially holds when $k \geq M$. In addition, when $k = 1$, this property reduces to the global disjoint projection assumption [26], which has been described as rather artificial in prior works [26, 2]. For this reason, we focus on the non-trivial regime by additionally requiring $1 < k < M$.

If we assume this property, we can guarantee that each membership tag contains at most k indices, thus $O(k \log M)$ bits suffice to serialize the membership tags². This makes the communication of $O(kN \log M)$ bits suffice in realizing $\mathcal{F}_{\text{FPSI-ME}}$, thereby achieving subquadratic communication overhead. In particular, the term $\log M$ is absorbed in the size of the underlying finite field of OKVS in our construction, i.e., $O(kN)$ communications of field elements suffice.

Relationship with Separation-Based Assumptions Unlike separation-based assumptions that explicitly enforce separability, the new assumption enforces bounded density, i.e., overlaps on δ -neighborhoods may happen. This discrepancy makes the k -max collision assumption incomparable with separation-based ones, as we visualized in Fig. 3.

In particular, recall that we can view the global disjoint projection assumption as a 1-max collision assumption. On the other hand, our assumption always holds whenever $k \geq M$, regardless of the structure of the data. From this perspective, we can interpret the k -max collision assumption as a middle-ground between the above-mentioned two configurations in terms of axis-wise density.

² This can be derived from a simple counting argument; $\sum_{i=0}^k \binom{M}{i} = O(M^k)$.

5.2 Plausibility Analysis

We provide the plausibility analysis on the proposed k -max collision. Similar to analyses on disjoint projection assumptions [2,23,43], we prove that the k -max collision holds on the set of uniformly sampled items over the universe with a high probability for a carefully chosen parameter k .

Theorem 2. *Let $\mathcal{U} := [0, 2^u) \cap \mathbb{Z}$ be the universe and $d \in \mathbb{N}$ be the data dimension. For a set X of size M whose elements are i.i.d. sampled from $\mathcal{U}^d$, the following inequality holds for $\mu = \frac{M \cdot (2\delta+1)}{2^u}$ and $k > \mu$:*

$$\Pr[X \text{ has a } k\text{-max collision property}] \geq 1 - d2^u \cdot e^{-k(\ln k - 1 - \ln \mu) - \mu}. \quad (3)$$

In particular, if $k \geq e\mu$, then the bound is further simplified by $1 - d2^u \cdot \left(\frac{e\mu}{k}\right)^k$.

Proof. Since intersections in one axis do not affect other axes, we can simplify the setting by considering each axis separately. That is, if we denote E_i as an event that the number of overlaps in δ -neighborhoods exceeds k at i^{th} axis, then we have that $1 - p^* \leq d \cdot \Pr[E_i]$ holds by the union bound, where p^* denotes the probability that X satisfies k -max collision property.

For $v \in \mathcal{U}$, define the random variable U_v as the number of elements $x \in X$ such that $v \in [x_i - \delta, x_i + \delta] \cap \mathbb{Z}$. Then U_v follows the binomial distribution $\text{Bin}(M, \frac{2\delta+1}{2^u})$. In particular, for any $v \in ([0, \delta) \cup (2^u - \delta, 2^u)) \cap \mathbb{Z}$, observe that $[v - \delta, v + \delta] \cap \mathcal{U} \subsetneq [v - \delta, v + \delta] \cap \mathbb{Z}$ holds. This implies the following inequality:

$$\Pr[E_i] \leq \Pr[\max_{v \in \mathcal{U}} U_v > k] \leq \sum_{v \in \mathcal{U}} \Pr[U_v > k] \leq 2^u \cdot \Pr[U_v > k].$$

Since U_v follows $\text{Bin}(M, \frac{2\delta+1}{2^u})$, we can utilize the Chernoff bound in Lemma 1, obtaining $\Pr[U_v > k] \leq \frac{e^{\mu\xi}}{(1+\xi)^{\mu(1+\xi)}}$, for $\xi = k\mu^{-1} - 1$. We can simplify this by

$$\Pr[U_v > k] \leq \frac{e^{\mu\xi}}{(1+\xi)^{\mu(1+\xi)}} = \frac{e^{k-\mu}}{\left(\frac{k}{\mu}\right)^k} = \mu^k \cdot e^{k-\mu} \cdot k^{-k} = \mu^k e^{-\mu} \cdot e^{-k(\ln k - 1)}.$$

This implies $1 - p^* \leq d \cdot \Pr[E_i] \leq d2^u \cdot \mu^k \cdot e^{-k(\ln k - 1) - \mu}$. The bound for $k \geq e\mu$ can be obtained from $\Pr[U_v > k] \leq \left(\frac{e\mu}{k}\right)^k$ by applying Lemma 1 for $k \geq e\mu$. $\square$

From this result, we can derive the following useful inequalities to capture the relationship between k and other parameters in terms of failure probability.

Corollary 1. *For the statistical security parameter κ and $\mu = \frac{M(2\delta+1)}{2^u}$, let us denote $A := \kappa + u + \log d$ and $\hat{A} := A + \mu \log e$. For the Lambert W function of branch $x \geq -e^{-1}$, $k \geq \frac{\hat{A}}{W(\hat{A}\mu^{-1}e^{-1})}$ ensures that k -max collision assumption holds with the probability $\geq 1 - 2^{-\kappa}$ on the set X in Theorem 2. Moreover, this bound can be simplified by $k \geq \max\{e\mu, \frac{A}{\log A - \log \mu - e}\}$ whenever $A \geq \frac{e^2 \mu}{\ln 2}$.*

Proof. See Appendix A.1. $\square$

Since $A = O(\log d)$, k scales with $\frac{\log d}{\log \log d}$ for a fixed κ, M, u . This indicates that the k -max collision property can handle high-dimensional data with a $\frac{\log d}{\log \log d}$ -growth in k . Furthermore, the failure probability scales linearly in d , i.e., it decreases for low-dimensional data. Hence, the k -max collision assumption can accommodate both low- and high-dimensional data with a slight adjustment on k , offering flexible parametrization on the dimension.

Interpretation of the Parameter μ We note that the parameter μ represents the local density of the set items per axis. In particular, in disjoint projection assumptions, $\mu < 1$ is necessary to avoid inherent overlaps derived by the pigeonhole principle; each item occupies $2\delta + 1$ points on $\mathcal{U}$ per axis, and $\mu \geq 1$ implies that there must be an overlapping point among occupied ones. On the other hand, since overlaps up to k are permitted, the k -max collision assumption is still valid for $\mu \geq 1$. This also shows that k -max collision supports a more flexible parameter regime than disjoint projection assumptions in terms of μ .

Furthermore, the role of μ can be interpreted more generally beyond the uniform distribution setting. Let us assume that the i^{th} coordinate of the data follows some distribution whose support is $\mathcal{U}$. If we denote X as the corresponding random variable, then for $p(v) := \Pr[|X - v| \leq \delta]$, U_v of the proof can be written as the summation of i.i.d. random variables $B_1, \dots, B_M$ that follows the Bernoulli distribution $\text{Ber}(p(v))$, i.e., U_v follows $\text{Bin}(M, p(v))$. That is, by using the same argument, we can obtain the same bound as Theorem 2 except for replacing μ with $M \cdot \max_{v \in \mathcal{U}} p(v)$. In this sense, the uniform data distribution serves as a special case with a constant local density $\frac{2\delta+1}{2^u}$.

Additional Remarks While we prove that the uniformly sampled set items satisfy the k -max collision property, similar to the disjoint projection assumptions, such an analysis may not align with real-world data. Nevertheless, like prior separation-based design paradigms, our assumption can also be locally verified on the given data before invoking the protocol. Using radix sort, one can count the number of overlaps for each axis in time $O(udM)$, and the detailed algorithm is provided in Appendix B.

In particular, our new assumption reflects how each axis discriminates between items from each other in feature-level sparsity, like feature-level discriminant analysis in data mining and pattern recognition [32]. For example, in the worst-case scenario where all the items overlap in a single axis (e.g., Fig. 3c), one may regard that axis as less useful for distinguishing items. Such a pruning is not inherent to separation-based designs, because they enforce sample-wise separability rather than feature-level discrimination on each coordinate.

5.3 Analysis on Concrete Parameters

Based on the result in Theorem 2, we analyze the desired regime of k under various parameter settings of $(d, M, u, \delta, \kappa)$. We sweep d and $\mu := \frac{M(2\delta+1)}{2^u}$ to report their effect on the concrete value of k . Unless otherwise stated, we fix $\kappa = 40$ and $u = 32$. In addition, we utilize the simplified bound for $k \geq e\mu$, because this condition holds throughout our experiment even when $k = 1$.

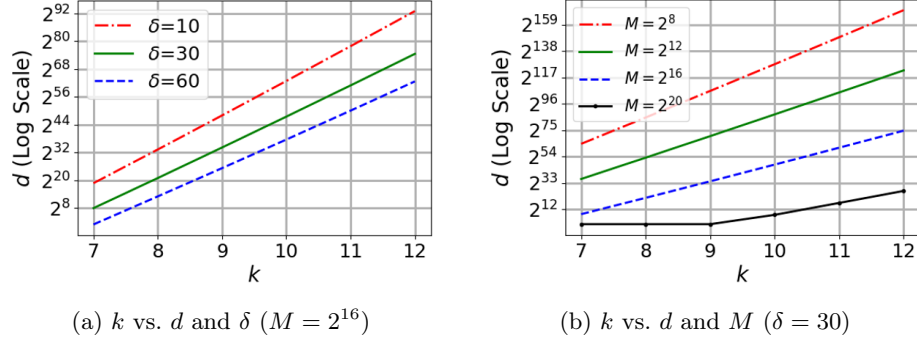


Fig. 6: Sensitivity analyses on k in δ (left) and M (right). A flat line in $M = 2^{20}$ of (b) means that no $d \geq 2$ satisfies failure probability less than 2^{-40} .

Sensitivity Analysis on d and δ In this experiment, we fix $M = 2^{16}$, which is often considered the largest experimental in prior FPSI studies. We consider the common presents of δ , namely, $\delta = 10, 30, 60$. For each δ value, we provide the maximum d achievable by the given k value. We provide the visualization result in Fig. 6a. As we expected in Theorem 2 and Corollary 1, we can observe that d exponentially scales with k . In particular, when $k = 10$, k -max collision assumption holds for $d \leq 2^{32}$ across all the choices of δ .

Sensitivity Analysis on d and M On the other hand, we provide a sensitive analysis of M , which takes We sweep $M \in \{2^8, 2^{12}, 2^{16}, 2^{20}\}$ and measure the same quantity as the previous experiment, while fixing $\delta = 30$. The result is visualized in Fig. 6b. We can observe the same tendency as in the previous analysis. The maximum d that satisfies failure probability $\leq 2^{-40}$ drastically decreases as M increases. Notably, when $M = 2^{20}$, relatively small k values (e.g., 7–9) do not ensure failure probability less than 2^{-40} . Nevertheless, larger k values, e.g., 10 or 11, provide a sufficient supporting dimension regime.

Concrete Ranges of d for k The Fig. 6a and 6b describe the asymptotic behavior well, showing that small k values are sufficient and can be regarded as a constant when $d < 10^3$. In addition to this, we provide a concrete range of d for the smallest k that achieves a failure probability $\leq 2^{-40}$. The results are provided in Tab. 2. Notably, this table indicates that, when $(M, \delta) = (2^{16}, 30)$, $k = 7$ is sufficient to handle $d \leq 262$, and for all parameter setups, $k = 11$ is sufficient to handle extremely large case $M = 2^{20}$ and $\delta = 60$.

Table 2: Ranges of d for parameters that achieve failure probability $\leq 2^{-40}$.

Params. (M, δ, k)	Supported d	Params. (M, δ, k)	Supported d
$(2^8, 10, 4)$	$d \leq 404$	$(2^{16}, 10, 7)$	$d \leq 458447$
$(2^8, 30, 5)$	$d \leq 7017268$	$(2^{16}, 30, 7)$	$d \leq 262$
$(2^8, 60, 5)$	$d \leq 225802$	$(2^{16}, 60, 8)$	$d \leq 8825$
$(2^{12}, 10, 5)$	$d \leq 1383$	$(2^{20}, 10, 9)$	$d \leq 4136$
$(2^{12}, 30, 6)$	$d \leq 631831$	$(2^{20}, 30, 10)$	$d \leq 179$
$(2^{12}, 60, 6)$	$d \leq 10372$	$(2^{20}, 60, 11)$	$d \leq 67$

6 The Proposed Protocol with k -Max Collision

6.1 Instantiation of the Protocol

Before describing the full protocol, we first show our instantiation of algebraic membership tag serialization via polynomials. We then show that, under the k -max collision assumption, $\mathcal{F}_{\text{FPSI-ME}}$ can be realized under the semi-honest model via an OKVS and an OKVS-compatible, ciphertext simulatable AHE.

Algebraic Membership Tags from Polynomials We consider the following polynomial serialization (Ser , DeSer , Inter) for a finite field $\mathbb{F}$, which is motivated by the polynomial-based set encoding framework by Kissner and Song [31].

- $\text{Ser} : S \in \{T \in \mathcal{P}([M]) : |T| \leq k\} \mapsto (f_1, \dots, f_{k+1}) \in \mathbb{F}^{k+1}$ such that for $\zeta(x) \stackrel{\$}{\leftarrow} \mathbb{F}[x]^{\leq k-|S|}$, $\sum_{i=1}^{k+1} f_i x^{i-1} = \prod_{z \in S} (x - z) \cdot \zeta(x)$ holds.
- $\text{DeSer} : (g_1, \dots, g_{2k}) \in \mathbb{F}^{\leq 2k} \mapsto \{j \in [M] : \sum_{i=1}^{k+1} g_i j^{i-1} = 0\}$.
- $\text{Inter} : (\mathbf{f}_{S_i})_{i=1}^d \in \prod_{i=1}^d \mathbb{F}^{(k+1)} \mapsto \sum \mathbf{r}_i \star \mathbf{f}_{S_i}$ for $\mathbf{r}_i \stackrel{\$}{\leftarrow} \mathbb{F}^k$, $i \in [d]$, where $\star$ is defined as follows: for the scalar product $\odot$,

$$(a_1, \dots, a_n) \star (b_1, \dots, b_m) := \sum_{j=1}^n a_j \odot (\underbrace{0, \dots, 0}_{j-1}, b_1, \dots, b_m, \underbrace{0, \dots, 0}_{n-j}) \in \mathbb{F}^{n+m-1}.$$

Note that the “ $\star$ ” operator corresponds to $(\sum_{i=1}^n a_i x^{i-1})(\sum_{j=1}^m b_j x^{j-1})$. In addition, with the k -max collision assumption, the degree of the serialized polynomial is bounded by k . This polynomial-based serialization indeed satisfies the desired properties mentioned in Section 4.

Lemma 3. *If $\frac{1}{|\mathbb{F}|} = \text{negl}(\kappa)$, then the polynomial set serialization (Ser , DeSer , Inter) defined as above satisfies serialization correctness, intersection correctness, and intersection obliviousness.*

Proof. See Appendix A.2. □

The Proposed Protocol We will use the following notation for cryptographic primitives needed for instantiating our protocol:

- OKVS : as an OKVS defined over $(\mathcal{K}, \mathcal{V})$.
- AHE : as an AHE defined over $(\mathcal{SK}, \mathbb{F}, \mathcal{C})$. AHE is κ -compatible with OKVS with an expansion rate $l \in \mathbb{N}$, i.e., there are mappings $\text{C2V} : \mathcal{C} \rightarrow \mathcal{V}^l$ and $\text{V2C} : \mathcal{V}^l \rightarrow \mathcal{C}$ satisfying properties in Definition 4.
- (Ser , DeSer , Inter) : the polynomial serialization mentioned above.

In addition, we consider a circuit $\mathcal{C} : \mathbb{F}^{d(k+1)} \times \mathbb{F}^{d(k-1)} \rightarrow \mathbb{F}^{2k}$ defined as

$$\begin{aligned} & \mathcal{C}((f_{j,m})_{j \in [d], m \in [k+1]}, (\gamma_{j,n})_{j \in [d], n \in [k]}) \\ &= \sum_{j=1}^d \sum_{n=1}^k \gamma_{j,n} \odot (\underbrace{0, \dots, 0}_{n-1}, f_{j,1}, \dots, f_{j,k+1}, \underbrace{0, \dots, 0}_{k-n}) \in \mathbb{F}^{2k} \quad (4) \end{aligned}$$

Parameters: Two parties, the receiver $\mathcal{R}$ and the sender $\mathcal{S}$. The universe set $\mathcal{U}$ and the data dimensionality $d \in \mathbb{N}$. The threshold $\delta > 0$. A collision-resistant hash function $H : \{0, 1\}^* \rightarrow \mathcal{K}$.

Inputs:

- $\mathcal{R}$: A set $X = \{x_1, \dots, x_M\} \subset \mathcal{U}^d$ of cardinality M .
- $\mathcal{S}$: A set $Y = \{y_1, \dots, y_N\} \subset \mathcal{U}^d$ of cardinality N .

Step 1 ($\mathcal{R}$): Encode and Serialize the Set

- Setup an AHE key $\text{sk} \leftarrow \text{AHE.Setup}(1^\lambda)$ and for $j \in [d]$, initialize key-value tables $M_j : \mathcal{U} \rightarrow \mathcal{P}([M])$ as empty sets. Initialize a list $S \leftarrow \emptyset$.
- For each $x_i \in X$ and $j \in [d]$ do:
 - For each $v \in [x_{i,j} - \delta, x_{i,j} + \delta] \cap \mathcal{U}$, look up $S_{v,j} \leftarrow M_j[v]$.
 - Update $S_{v,j} \leftarrow S_{v,j} \cup \{i\}$.
- For each $j \in [d]$ and $v \in \mathcal{U}$ such that $M_j[v] \neq \emptyset$,
 - Set $(f_m)_{m \in [k+1]} \leftarrow \text{Ser}(M_j[v])$, $(g_{m,n})_{n \in [l]} \leftarrow \text{C2V}(\text{AHE.Enc}(\text{sk}, f_i))$.
 - Append $S \leftarrow S \cup \{(H(v \| j \| m \| n), g_{m,n})\}$ for $m \in [k+1]$, $n \in [l]$.
- If $|S| < (2\delta + 1)(k+1)ldM$, append $S \leftarrow S \cup \{H((\gamma_1 \| 0 \| 0 \| 0), \gamma_2)\}$ for $\gamma_1 \xleftarrow{\$} \mathcal{U}$ and $\gamma_2 \xleftarrow{\$} \mathbb{F}$ to ensure $|S| = (2\delta + 1)(k+1)ldM$.
- Construct $\text{okvs} \leftarrow \text{OKVS.Encode}(S)$ and send okvs to $\mathcal{S}$.

Step 2 ($\mathcal{S}$): Obviously Compute Membership Tag Intersection

- Upon receiving okvs from $\mathcal{R}$, for each $y_i \in Y$ and $j \in [d]$ do:
 - Set $c_{i,j,m} \leftarrow (\text{OKVS.Decode}(\text{okvs}, H(y_{i,j} \| j \| m \| n)))_{n \in [l]} \in \mathbb{F}^l$, $m \in [k+1]$.
 - Set $(\text{ct}_{i,j,m})_{m \in [k+1]} \leftarrow (\text{V2C}(c_{i,j,m}))_{m \in [k+1]} \in \mathcal{C}^{k+1}$.
 - Sample $\gamma_{i,j,n} \leftarrow \mathbb{F}^\times$ for $n \in [k]$.
- Set $\hat{\text{ct}}_i \leftarrow \text{Eval}(\mathcal{C}, (\text{ct}_{i,j,m})_{j \in [d], m \in [k+1]}, (\gamma_{i,j,n})_{j \in [d], n \in [k]}) \in \mathcal{C}^{2k}$, $i \in [N]$.
- Send $\{\hat{\text{ct}}_i\}_{i \in [N]}$ to $\mathcal{R}$ with indices shuffled.

Step 3 ($\mathcal{R}$): Interpret the Result

- Upon receiving $S \in \mathcal{C}^{N \times 2k}$ from $\mathcal{R}$, initialize a set $R \leftarrow \emptyset$.
- For each $(\hat{\text{ct}}_{i,1}, \dots, \hat{\text{ct}}_{i,2k}) \in \mathcal{C}^{2k}$ in S :
 - Compute $\hat{f}_i \leftarrow (\text{AHE.Dec}(\text{sk}, \hat{\text{ct}}_{i,j}))_{j \in [2k]} \in \mathbb{F}^{2k}$.
 - Retrieve $S_i \leftarrow \text{DeSer}(\hat{f}_i) \in \mathcal{P}([M])$ and append $R \leftarrow R \cup \{S_i\}$.

Return R .

Fig. 7: The Proposed Protocol Π_{FPSI} .

where $\odot$ denotes scalar multiplication. Recall that this circuit is equivalent to compute $\sum_{j=1}^d f_j(x) \gamma_j(x)$ where $f_j(x) = \sum_{m=1}^{k+1} f_{j,m} x^{m-1} \in \mathbb{F}[x]^{\leq k}$ and $\gamma_j(x) = \sum_{n=1}^k \gamma_{j,n} x^{n-1} \in \mathbb{F}[x]^{\leq k-1}$ in terms of coefficients, i.e., if we denote a vector $\mathbf{x} = (1, x, \dots, x^{2k-1})$ of an indeterminate x and the output of $\mathcal{C}$ as $\mathbf{c}$, then $\langle \mathbf{c}, \mathbf{x} \rangle = \sum_{j=1}^d f_j(x) \gamma_j(x)$. Note that evaluating $\mathcal{C}$ requires $O(k^2 d)$ field operations.

Recall that we denote computational and statistical security parameters as λ and κ , respectively, and we consider $\mathcal{U} = [0, 2^u) \cap \mathbb{Z}$ as the universe. In addition, we denote a collision-resistant hash function $H : \{0, 1\}^* \rightarrow \mathcal{K}$ and $\cdot \| \cdot$ as a bit-wise concatenation. We present the proposed protocol Π_{FPSI} in Fig. 7.

When constructing an OKVS, we remark that the number of encoded key-value pairs would be less than $(2\delta + 1)(k + 1)ldM$, where $(k + 1)l$ stems from the expansions from C2V and Ser. To avoid the information leakage from the OKVS encoding size, we add dummy key-value pairs with the domain separation. In addition, from the protocol output $R = \{S_i\}_{i \in [N]}$ of the $\mathcal{R}$, the bipartite graph defined as in $\mathcal{F}_{\text{FPSI}}$ can be built by computing the set R_i for $i \in [M]$ that consists of $j \in R_i$ iff $i \in S_j$. Since the index set $[N]$ is randomly permuted by $\mathcal{S}$, $\mathcal{R}$ learns the incidence relation only, without learning the exact matched items of $\mathcal{S}$.

From the implementation perspective, we assume that the key-value tables M_j support constant-time look up (in average time), likewise hash table data structure. In addition, for memory efficiency, $\mathcal{R}$ can dynamically update the non-empty indices, instead of pre-fetching a memory of size $O(|\mathcal{U}| \cdot kd)$ bits.

6.2 Analysis of the Protocol

We now provide the security and complexity analyses of the proposed protocol.

Security Proof The high-level roadmap of the security proof is to show that (i) $\mathcal{R}$'s view can be simulated by OKVS independence and OKVS compatibility, and (ii) $\mathcal{S}$'s view can be simulated by the ciphertext simulatability and the design of the membership tag serialization. Recall that, by Lemma 2, it suffices to realize $\mathcal{F}_{\text{FPSI-ME}}$ instead of $\mathcal{F}_{\text{FPSI}}$. We provide the formal proof in Theorem 3.

Theorem 3. *If the following properties hold and $\mathcal{R}$'s set has the k -max collision property, then Π_{FPSI} securely realizes $\mathcal{F}_{\text{FPSI}}$ for $L := (2\delta + 1)(k + 1)ldM$:*

- OKVS satisfies properties in Definition 1 and 2 (Independence).
- AHE satisfies properties in Definition 3 (Ciphertext Simulatability) and 4 (OKVS Compatibility).
- H is collision-resistant and $|\mathcal{K}| \geq 2^{\kappa + \log L + \log(L + dN)}$.
- $M, N = \text{poly}(\kappa)$ and $\frac{1}{|\mathbb{F}|} = \text{negl}(\kappa)$.

Proof. We prove the security of the protocol in two parts: (i) whether the protocol returns the same output as the ideal functionality (Correctness), and (ii) the existence of the simulator (Security).

(Correctness) To characterize the output distribution of the protocol, we consider the following case analysis.

- Case 1: We first consider the case when OKVS keys collide with each other. This can happen with two failure modes, namely, (i) $\mathcal{R}$'s OKVS encoding key collides, and (ii) $\mathcal{S}$ unexpectedly decodes a valid value from a key collision. We analyze each case separately:
 - Case 1-1: Since $\mathcal{R}$ encodes $L := (2\delta + 1)(k + 1)ldM$ key-value pairs during OKVS encoding, the birthday bound deduces that the probability of this case happening is $\frac{L^2}{|\mathcal{K}|}$.

- Case 1-2: Since $\mathcal{S}$ tries dN decoding of its key on the retrieved OKVS, the probability for this case happening can be obtained by the union bound, namely, $\frac{dLN}{|\mathcal{K}|}$.

To sum up, the probability of Case 1 is bounded by $\frac{L^2+dLN}{|\mathcal{K}|}$, which is negligible in κ when $|\mathcal{K}| \geq 2^{\kappa+\log L+\log(L+dN)}$.

- Case 2: While the OKVS key collision may not happen, the failure case may happen because of the polynomial serialization. Note that there are no false rejections, i.e., matching items are always marked as matched in Π_{FPSI} . On the other hand, false positives, i.e., non-matching items that are marked as matched, may happen. For the intersection S corresponding to a $\mathcal{S}$'s item and $w \in [M] \setminus S$, we claim that the probability that Π_{FPSI} determines w as a matched item is at most $\frac{2}{|\mathbb{F}|}$.

Proof (Proof of Claim). Let $\hat{f}(x)$ be a polynomial retrieved by $\mathcal{R}$. It suffices to show that $\Pr[\hat{f}(w) = 0] = \frac{1}{|\mathbb{F}|}$. If $\mathcal{S}$ decoded the OKVS with a key that was not previously encoded, say i^{th} coordinate, then for some polynomial $g(x) \in \mathbb{F}[x]^{\leq d}$ and $\gamma_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq t}$, $\hat{f}(w)$ is blinded by $g(w)\gamma_i(w)$. Here, $g(x)$ is indistinguishable from a random polynomial due to the independence of an OKVS³, so $g(w)\gamma_i(w)$ can be regarded as a product of two random variables over $\mathbb{F}$. This implies that $\Pr[g(w)\gamma_i(w) = 0] = \frac{2}{|\mathbb{F}|}$.

In contrast, if $\mathcal{S}$ correctly decodes all the keys, but $w \notin S$, then we can select at least one S_i such that $w \notin S_i$. Hence, the distribution of $\prod_{z \in S_i} (w - z)\zeta_j(w)\gamma_j(w)$ is uniformly random over $\mathbb{F}$, so is $\hat{f}(w)$.

For both cases, we obtain the false positive probability of at most $\frac{2}{|\mathbb{F}|}$, which completes the proof. $\square$

Hence, by applying the union bound on at most M candidates and N items held by $\mathcal{S}$, we obtain the upper bound of the failure probability by $\frac{2MN}{|\mathbb{F}|}$.

From these case analyses, we can conclude that, under the selected parameters, the probability that one of Case 1 or Case 2 happens is negligible in κ , which proves the correctness of Π_{FPSI} .

(Security) Since each party's internal randomness is modeled as a public coin, we focus on simulating the transcripts exchanged during the protocol.

For corrupted $\mathcal{S}$, the simulator $\text{Sim}_{\mathcal{S}}$ must simulate the received **okvs**. To this end, $\text{Sim}_{\mathcal{S}}$ first constructs key-value pairs $\{(k_i, v_i) : k_i \xleftarrow{\$} \mathcal{K}, v_i \xleftarrow{\$} v\}$ of size $(2\delta + 1)(k + 1)ldM$ and return the OKVS encoding, say $\widehat{\text{okvs}}$ of these pairs. We claim that **okvs** and $\widehat{\text{okvs}}$ are computationally indistinguishable in the security parameter λ . To show this, we consider the following hybrid argument.

- Hyb₀: This case corresponds to the real protocol execution.

³ Recall that the probability runs over the internal randomness of OKVS.Encode .

- **Hyb₁**: $\mathcal{R}$ replaces each OKVS value to one randomly sampled $v \xleftarrow{\$} \mathcal{V}$, instead of ciphertexts with C2V. Hence, the advantage of distinguishing **Hyb₀** and **Hyb₁** is bounded by that derived from the OKVS compatibility (**obliviousness** in Definition 4), which is $\text{negl}(\lambda)$.
- **Hyb₂**: The OKVS decoding is replaced by the following procedure: if an OKVS key k was previously encoded in **Hyb₁**, then return the programmed value; otherwise, return a fresh random element $v \xleftarrow{\$} \mathcal{V}$. We treat decoding as part of $\mathcal{S}$'s view and define this hybrid by modifying the distribution of decoding outputs on non-programmed keys. Since $\mathcal{S}$ tries polynomially many OKVS decoding in λ , by the independence of the OKVS, the advantage of distinguishing **Hyb₁** and **Hyb₂** is negligible in $\text{negl}(\lambda)$.
- **Hyb₃**: This case corresponds to the simulator's output. Compared to **Hyb₂**, the difference comes from the choice of OKVS keys. Since both hybrids encode the same number of key-value pairs, by the obliviousness of the OKVS, the advantage of distinguishing **Hyb₂** and **Hyb₃** is bounded by $\text{negl}(\lambda)$.

Therefore, the corrupted $\mathcal{S}$'s view is indistinguishable from $\text{Sim}_{\mathcal{S}}$'s output.

On the other hand, for corrupted $\mathcal{R}$, the simulator $\text{Sim}_{\mathcal{R}}$ must simulate the received ciphertexts $\{\hat{\text{ct}}_i\}_{i=1}^N$ from the ideal functionalities' output. For simplicity, we denote $S_i \subset [M]$ as the indices of $\mathcal{R}$'s item matched from $\hat{\text{ct}}_i$. We denote the simulator $\text{Sim}_{\mathcal{C}}$ for the inner product circuit of length $k+1$. We focus on the case when $\mathcal{R}$ receives a non-trivial matching; if there were no intersection, then the opened $\mathcal{S}$ is indistinguishable from a random string due to the independence of an OKVS. We consider the following hybrid argument to design the $\text{Sim}_{\mathcal{R}}$:

- **Hyb₀**: This case corresponds to the real protocol execution.
- **Hyb₁**: $\mathcal{S}$'s internal computation and output ciphertext are modified as follows. Instead of opening the OKVS received from $\mathcal{R}$, for each S_i , $i \in [N]$, and each $j \in [d]$, $m \in [k+1]$, compute $f_{i,j}(x) \leftarrow \prod_{z \in S_i} (x - z) \cdot \zeta_{i,j}(x)$ for $\zeta_{i,j}(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k-|S_i|}$ and set $\text{ct}_{i,j,m}$ as the encryption of the coefficient of $f_{i,j}(x)$ on x^m . Then for randomly sampled $\gamma_{i,j}(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k-1}$, homomorphically compute $\sum_{j=1}^d f_{i,j}(x)\gamma_{i,j}(x)$ with coefficients $\gamma_{i,j}(x)$ and each $\text{ct}_{i,j,m}$, obtaining the ciphertexts $\hat{\text{ct}}_1, \dots, \hat{\text{ct}}_{2k}$. The difference between **Hyb₀** and **Hyb₁** is characterized by (**Intersection Obliviousness**) of the polynomial membership tags. **Hyb₁** compiles the simulator corresponding to this property, except for the fact that it evaluates this computation over the ciphertexts. Since $\mathcal{R}$ has an AHE secret key, and the same circuit is homomorphically executed in both **Hyb₀** and **Hyb₁**, its view in terms of $\mathcal{R}$ is determined by the underlying plaintext. Therefore, by Lemma 3, the advantage of distinguishing **Hyb₀** and **Hyb₁** is $\text{negl}(\kappa)$.
- **Hyb₂**: We replace the ciphertexts $\{\hat{\text{ct}}_i\}_{i \in [N]}$ with the output of

$$\text{Sim}_{\mathcal{C}}(\mathcal{C}, (\{\text{ct}_{i,j,m}\}_{j \in [d], m \in [k+1]}), (\{\gamma_{i,j,l}\}_{j \in [d], l \in [t+1]})),$$

where $\mathcal{C}$ is the circuit to compute $\sum_{j=1}^d f_j(x)\gamma_j(x)$ as defined in Eq. (4), $\text{ct}_{j,m}$ is the same as **Hyb₁**, and $\gamma_{j,l} \xleftarrow{\$} \mathbb{F}$. Since the given AHE is ciphertext

simulatable, we can ensure that the advantage of distinguishing Hyb_1 and Hyb_2 is bounded by $\text{negl}(\lambda)$.

Hence, if we define $\text{Sim}_{\mathcal{R}}$ as the simulation output of $\text{Sim}_{\mathcal{C}}$ in Hyb_2 , we can deduce that $\mathcal{R}$'s view is computationally indistinguishable from $\text{Sim}_{\mathcal{R}}$'s output with an advantage at most $\text{negl}(\lambda) + \text{negl}(\kappa)$.

To conclude, the proposed protocol securely realizes $\mathcal{F}_{\text{FPSI}}$ in the semi-honest model, thereby completing the proof. $\square$

Efficiency Analysis We present the efficiency analysis of Π_{FPSI} . We count the number of field operations and AHE ciphertext operations. We assume that OKVS encoding and decoding take $O(L\kappa)$ and $O(\kappa)$ field operations for L key-valued pairs, respectively [5]. We denote ϵ as the expansion rate of an OKVS, i.e., L key-value pairs are encoded to $(1+\epsilon)L$ field elements. We assume that the polynomial factorization of degree k over a field $\mathbb{F}$ takes $k \log^2 k (\log |\mathbb{F}| + \log k)$ field operations [41, Corollary 14.16], and the cost for computing Ser as $k \log^2 k$.

Theorem 4. Π_{FPSI} exhibits computation/communication costs as follows. For simplicity, we denote $L = (2\delta + 1)(k + 1)ldM$

- (Communication) $(1 + \epsilon)L \cdot \mathbb{F} + (k + 1)N \cdot \text{ct}$
- ($\mathcal{R}$'s Computation) $(k \log^2 k \cdot (2\delta + 1)dM + \kappa L + (2k \log^2 2k + 2k \log^2 2k \log |\mathbb{F}|)N) \cdot \mathbb{F}_{\text{Mul}} + L \cdot \text{ct}_{\text{Enc}} + 2kN \cdot \text{ct}_{\text{Dec}}$
- ($\mathcal{S}$'s Computation) $\kappa dkN \cdot \mathbb{F} + dk^2N \cdot \text{ct}_{\text{SMul}} + (d - 1)k^2N \cdot \text{ct}_{\text{Add}}$

where $\mathbb{F}$ and ct denote the size of a field element and an AHE ciphertext, respectively. $\mathbb{F}_{\text{Mul}}$, ct_{Add} , ct_{SMul} , ct_{Enc} , ct_{Dec} denote field multiplication, AHE addition, scalar multiplication, encryption, and decryption, respectively.

Proof. We analyze each item one by one.

- (Communication) The communication appears when sending okvs and retrieving $\{\hat{\text{ct}}_i\}_{i \in [N]} \in \mathcal{C}^{N \times (k+1)}$. Since okvs encodes L key-value pairs, it requires $(1 + \epsilon)L$ field elements to represent. This verifies the claim.
- ($\mathcal{R}$'s Computation) We can break down the computation into three parts:
 - (Serialization and Encryption): $\mathcal{R}$ needs to serialize at most $(2\delta + 1)dM$ sets, which takes $k \log^2 k \cdot (2\delta + 1)dM$ field operations. After serialization, $\mathcal{R}$ needs to encrypt L ciphertexts.
 - (OKVS Encoding): To encode okvs , $\mathcal{R}$ requires $\kappa \cdot L$ field operations.
 - (Result Analysis): After retrieving $\{\hat{\text{ct}}_i\}_{i \in [N]} \in \mathcal{C}^{N \times 2k}$, $\mathcal{R}$ decrypts all of them and factorize the retrieved polynomials. These takes $2kN$ ciphertext decryptions and $(2k \log^2 2k + 2k \log^2 2k \log |\mathbb{F}|)N$ field operations, respectively.

By merging them, we obtain the desired result.

- ($\mathcal{S}$'s Computation) $\mathcal{S}$'s computation involves dN OKVS decodings and evaluating the circuit $\mathcal{C}$ in Eq. (4) N times. These takes κdkN field operations, $(d - 1)k^2N$ ciphertext additions, and dk^2N ciphertext scalar multiplications, respectively. $\square$

Overall, Π_{FPSI} achieves $O(kd\delta M + kN)$ communication, $O(k \log^2 k (\delta dM + N))$ $\mathcal{R}$'s computation, and $O(k^2 dN)$ $\mathcal{S}$'s computation complexities.

7 Concluding Remark and Future Direction

In this study, we present a new design paradigm for an efficient FPSI protocol based on the k -max collision assumption. Instead of pairwise separability as a proxy of sparsity, our new assumption captures the bounded density of data per axis, thereby intrinsically reflecting the axis-wise discriminative power. With our membership tag-based abstraction, our protocol achieves computation/communication complexities comparable to those of separation-based protocols—scaling linearly with each party’s set sizes and the data dimension—while covering the previously unsupported intermediate-dimension regime. Overall, our result demonstrates that prevailing separation-based paradigms may not be fundamental for achieving efficient FPSI.

We leave several interesting directions for future study. First, while we established the feasibility of k -max collision assumption in uniformly random data, further analyses under diverse data distributions would deepen our understanding. As we briefly discussed in Section 5.2, the local density parameter μ provides a unified lens for interpreting data distributions in terms of bounded collision. We expect that this density-bounded perspective can be naturally combined with statistical pattern recognition techniques based on discriminant analysis [32]. This motivates us to investigate the k -max collision assumption in similarity matching-related tasks, e.g., (approximate) nearest neighbor search [14]. In addition, following recent FPSI proposals, instantiating our framework via lightweight cryptographic primitives [36,3,43] instead of AHEs, together with optimizations such as prefix representation [13,17,3], would be an interesting direction for improving its concrete efficiency. Finally, extending our framework, e.g., by incorporating richer functionality, supporting additional metrics (e.g., $\ell_{p \in \mathbb{N}}$), or enhancing security against a malicious adversary, under the density-bounded assumption, would also be a promising direction.

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Appendix

A Omitted Proofs

We provide the proofs of lemmas, whose proofs were omitted in the main text.

A.1 Proof of Corollary 1

Proof. From the result of Theorem 2, we know that

$$\begin{aligned} d2^u \cdot \mu^k \cdot e^{-k(\ln k - 1) - \mu} &\leq 2^{-\kappa} \\ \Rightarrow \log d + u + k \log \mu + k \log e - k \log k + \mu \log e &\leq -\kappa \\ \Rightarrow \kappa + u + \log d + \mu \log e &\leq k(\log k - \log \mu - \log e), \end{aligned}$$

From this, if we substitute $\hat{A} = \kappa + u + \log d + k \log e$, $B := \log \mu + \log e$ and $\omega = \ln k$, we have that $\log k = \frac{\omega}{\ln 2}$, and

$$k(\log k - \log \mu - \log e) = e^\omega \left(\frac{\omega}{\ln 2} - \frac{\ln \mu + 1}{\ln 2} \right) = \frac{e\mu}{\ln 2} e^{\omega - \ln e\mu} (\omega - \ln e\mu)$$

That is, we obtain the inequality $e^{\omega - \ln e\mu} (\omega - \ln e\mu) \geq \frac{\hat{A} \ln 2}{e\mu}$.

From this, observe that $k > \mu$ implies $\omega - \ln e\mu > -1$. Hence, for the Lambert W function of branch $x \geq -e^{-1}$, we obtain

$$\begin{aligned} k &\geq \exp \left\{ W \left(\frac{\hat{A} \ln 2}{e\mu} \right) + \ln e\mu \right\} \\ &= \frac{e\mu}{W \left(\frac{\hat{A} \ln 2}{e\mu} \right)} \cdot W \left(\frac{\hat{A} \ln 2}{e\mu} \right) \exp \left\{ W \left(\frac{\hat{A} \ln 2}{e\mu} \right) \right\} = \frac{\hat{A} \ln 2}{W(\hat{A} \mu^{-1} e^{-1} \ln 2)}. \end{aligned}$$

Thus, by leveraging the inequality $W(z) \leq \ln z$, i.e., $\frac{1}{W(z)} \geq \frac{1}{\ln z}$ whenever $z \geq e$, under the condition that $\hat{A} \geq \frac{e^2 \mu}{\ln 2}$, we finally obtain

$$k \geq \frac{\hat{A} \ln 2}{\ln \hat{A} - \ln \mu - \ln e + \ln 2} = \frac{\hat{A}}{\log \hat{A} - \log \mu - \log e + 1}.$$

In particular, when $k \geq e\mu$, the term $\mu \log e$ is discarded, so we obtain the same bound except for replacing $\hat{A}$ with $A := \kappa + u + \log d$. $\square$

A.2 Proof of Lemma 3

While proving (**Serialization Correctness**) and (**Intersection Correctness**) are straightforward, proving (**Intersection Obliviousness**) requires an additional lemma to apply Kissner and Song's result [31, Lemma 2]. For this reason, we first provide the proof for the former two properties as follows:

Proof. (Serialization Correctness) If we denote $g(x) := \sum_{i=1}^{k+1} g_i x^{i-1}$ for $S \in \mathcal{P}([M]; k)$ and $(g_1, \dots, g_{k+1}) \leftarrow \text{Ser}(S)$, we have that $g(j) = 0$ if and only if (i) $j \in S$ or $\zeta(j) = 0$. That is, the event $\text{DeSer}(\text{Ser}(S)) \neq S$ happens only when $\zeta(j) = 0$ for some $j \notin S$. Since $\zeta(j)$ follows the uniform distribution over $\mathbb{F}$, we obtain that $\Pr[\text{DeSer}(\text{Ser}(S)) \neq S] \leq \frac{M-|S|}{|\mathbb{F}|} \leq \frac{M}{|\mathbb{F}|}$, which is negligible in κ .

(Intersection Correctness) For simplicity, let us denote $S = \bigcap_{i=1}^d S_i$ and $D_i := S_i \setminus S$. If we denote $f_{S_i} := \prod_{z \in S_i} (x - z) \cdot \zeta_i(x)$ and $\gamma_i(x) \in \mathbb{F}[x]^{\leq k-1}$ as the polynomial corresponding to $\mathbf{r}_i$, then the polynomial corresponding to $\sum_{i=1}^d \mathbf{r}_i \star \mathbf{f}_{S_i}$ can be calculated as

$$\prod_{z \in S} (x - z) \cdot \sum_{i=1}^d \left(\prod_{z \in D_i} (x - z) \cdot \zeta_i(x) \cdot \gamma_i(x) \right).$$

From this, we can observe that, if we denote the polynomial that lies inside the above parentheses as $\hat{\zeta}(x)$, the evaluation result $\hat{\zeta}(j)$ for $j \notin S$ follows the uniform distribution over $\mathbb{F}$, where the probability runs over the choice of $\gamma_i(x)$ and $\zeta_i(x)$. This is because, for the constant term $\zeta_{i,1}$ of $\zeta_i(x)$, such a distribution is blinded by the quantity $\gamma_i(j) \cdot \prod_{z \in D_i} (j - z) \cdot \zeta_{i,1}$ that follows the uniform distribution over $\mathbb{F}$. This completes the proof. $\square$

To prove **(Intersection Obliviousness)**, we introduce the following lemma, which ensures that multiplying a random polynomial largely preserves the co-prime relationship.

Lemma 4. *Let $f_1(x), \dots, f_d(x) \in \mathbb{F}[x]^{\leq k}$ be polynomials s.t. $\gcd(f_1, \dots, f_d) = 1$. For a randomly sampled $\zeta_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k - \deg f_i(x)}$, $i \in [d]$, the following inequality holds:*

$$\Pr[\gcd(\zeta_1 f_1, \dots, \zeta_d f_d) = 1] \geq 1 - \frac{kd}{|\mathbb{F}|}.$$

Proof. Suppose that $h(x) := \gcd(\zeta_1 f_1, \dots, \zeta_d f_d)$ for some non-constant polynomial $h(x)$. Then, since $\gcd(f_1, \dots, f_d) = 1$, we can select $i \in [d]$ such that $f_i(x)$ is not divided by $h(x)$. This implies that $h(x)$ must divide $\zeta_i(x)$. Since $\zeta_i(x)$ is a random polynomial of degree at most $k - \deg f_i$, we can deduce that

$$\Pr[h(x) \text{ divides } \zeta_i(x)] \leq |\mathbb{F}|^{-\deg h(x)} \leq |\mathbb{F}|^{-1}. \quad (5)$$

We can take a union bound with respect to indices i and the choice of $h(x)$. Note that, for each i , the number of irreducible factors of $\zeta_i(x)$ is at most k , and it suffices to count the number of such factors as candidates of $h(x)$. That is, we obtain the following upper bound:

$$\Pr[\gcd(\zeta_1 f_1, \dots, \zeta_d f_d) \neq 1] \leq \sum_{i=1}^d \frac{k}{|\mathbb{F}|} = \frac{kd}{|\mathbb{F}|}.$$

This completes the proof. $\square$

With this lemma, we prove the following statement, which directly implies the construction of the simulator for **Intersection Obliviousness**; we can regard the Exp_1 as the desired simulator, as it takes S only but its output polynomial is statistically close enough (with a margin by $\frac{2kd}{|\mathbb{F}|}$) with the output of **Inter**.

Lemma 5. *Let $S_1, \dots, S_d$ ($d \geq 2$) be fixed sets s.t. $|S_i| \leq k$. Define $S := \bigcap_{i=1}^d S_i$ and $\tilde{f}_{S_i}(x) := \prod_{z \in S_i} (x - z) \in \mathbb{F}[x]$. If $t \geq k - 1$, then the following two experiments, Exp_1 and Exp_2 , have statistical distance at most $\frac{2kd}{|\mathbb{F}|}$.*

$\text{Exp}_0(\tilde{f}_{S_1}(x), \dots, \tilde{f}_{S_d}(x))$:	$\text{Exp}_1(\tilde{f}_S(x))$:
– Sample $\zeta_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k- S_i }$.	– Sample $\zeta_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k- S }$.
– Sample $\gamma_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq t}$.	– Sample $\gamma_i(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq t}$.
– Output $T \leftarrow \sum_{i=1}^d \gamma_i(x) \tilde{f}_{S_i}(x) \zeta_i(x)$.	– Output $T \leftarrow \tilde{f}_S(x) \sum_{i=1}^d \gamma_i(x) \zeta_i(x)$.

Proof. We consider the following auxiliary experiment $\widehat{\text{Exp}}$, which takes $\tilde{f}_S(x)$ as input and output $\tilde{f}_S(x) \cdot h(x)$ for $h(x) \xleftarrow{\$} \mathbb{F}[x]^{\leq k-|S|+t}$.

Exp_0 and $\widehat{\text{Exp}}$: For simplicity, let $D_i := S_i \setminus S$. Then we can write

$$\sum_{i=1}^d \gamma_i(x) \tilde{f}_{S_i}(x) \zeta_i(x) = \tilde{f}_S(x) \cdot \sum_{i=1}^d \gamma_i(x) \tilde{f}_{D_i}(x) \zeta_i(x). \quad (6)$$

We can observe that $\gcd(\tilde{f}_{D_1}, \dots, \tilde{f}_{D_d}) = 1$, and $\deg \tilde{f}_{D_i}(x) \zeta_i(x) = k - |S|$. Thus, by Lemma 4, we can ensure that $\gcd(\tilde{f}_{D_1} \zeta_1, \dots, \tilde{f}_{D_d} \zeta_d) = 1$ except for the failure probability at most $\frac{(k-|S|)d}{|\mathbb{F}|} \leq \frac{kd}{|\mathbb{F}|}$. In addition, if we substitute $f_i(x) := \tilde{f}_{D_i}(x) \zeta_i(x)$, then we can re-write the R.H.S. of Eq. (6) as $\tilde{f}_S(x) \cdot \sum_{i=1}^d \gamma_i(x) f_i(x)$.

Here, we can utilize the Kissner and Song's result [31, Lemma 2] because $\gcd(f_1, \dots, f_D) = 1$, $\deg f_i \leq k - |S|$, and $\deg \gamma_i = t \geq k - 1$. This implies that the distribution of $\sum_{i=1}^d \gamma_i(x) f_i(x)$ is statistically identical to the uniform distribution over $\mathbb{F}[x]^{k-|S|+t}$. Thus, except for the case when $\gcd(\tilde{f}_{D_1} \zeta_1, \dots, \tilde{f}_{D_d} \zeta_d) \neq 1$, these two probability ensembles are indistinguishable, which implies that the statistical distance of them are at most $\frac{kd}{|\mathbb{F}|}$.

Exp_1 and $\widehat{\text{Exp}}$: We can think of Exp_1 as a special case Exp_0 when $\tilde{f}_{D_i} \equiv 1$ for all $i \in [d]$. In this case, we still can apply Lemma 4 to ensure that $\gcd(\zeta_1, \dots, \zeta_d) = 1$ with a failure probability of at most $\frac{kd}{|\mathbb{F}|}$, the remaining argument on applying Kissner and Song's result do not affect the choice of each $\tilde{f}_{D_i}$. Therefore, we obtain the same statistical distance as in Exp_0 and $\widehat{\text{Exp}}$.

To sum up, by merging the above results, we finally obtain that Exp_0 and Exp_1 are statistically indistinguishable with statistical distance at most $\frac{2kd}{|\mathbb{F}|}$. This completes the proof. $\square$

Therefore, the polynomial-based membership tag satisfies all the claimed properties, completing the proof. $\square$

B Local Verification of k -Max Collision Property

For a set $X \subset \mathcal{U}^d$ of size M , we provide an algorithm to verify the k -max collision property of X in time $O(udM)$. The code idea of the algorithm is to scan over the sorted coordinates; if we denote $x_{1,i}, \dots, x_{M,i}$ the sorted i^{th} coordinates of items of X , the maximum number of overlaps corresponds to $\max_{i>j} \{i - j + 1 : x_i - x_j \leq 2\delta\}$. The correctness of this approach is clear, because if we find endpoints of the intersecting regime, say x, y , then all the other points intersecting at that regime must lie in the interval $[x, y]$. We describe the detailed procedure in Algorithm 1.

Algorithm 1 Verification of k -Max Collision Property

Require: $X \subset \mathcal{U}^d$ such that $|X| = M$, threshold $\delta \in \mathbb{N}$, and $k \in \mathbb{N}$.
Ensure: TRUE or FALSE for whether X satisfies the k -max collision property.

```

1: for  $i$  from 1 to  $d$  do
2:   Sort and enumerate the  $i^{\text{th}}$  component of items  $X$ , say  $(x_{1,i}, \dots, x_{M,i}) \in \mathcal{U}^M$ .
3:   Initialize  $C \leftarrow 0$  and  $s \leftarrow 1$ .
4:   for  $r$  from 1 to  $M$  do
5:     while  $x_{r,i} - x_{s,i} > 2\delta$  do
6:       Update  $s \leftarrow s + 1$ .
7:     end while
8:     Update  $C \leftarrow \max\{C, r - s + 1\}$ 
9:     If  $C > k$ : return FALSE.
10:  end for
11: end for
12: return TRUE

```

Complexity Analysis The algorithm consists of two parts of subroutines: (i) sorting each component, and (ii) scanning j and k over the range from 1 to M . These subroutines are shared over d axes, so we provide the analysis for each of them. For the former, we can employ radix sort, which requires $O(uM)$ time complexity. On the other hand, for the latter, we can observe that j and k are being scanned over $[M]$, and they are monotonically increasing. In addition, the number of comparisons for checking $x_{r,i} - x_{s,i} \stackrel{?}{>} 2\delta$ and $C \stackrel{?}{>} r - s + 1$ are at most M each, hence $2M$ at most comparisons suffice. Therefore, aggregating over all d axes, Algorithm 1 ends within $O(udM)$ time complexity.